

Q.1)

Ans) b

Exp) Option b is the correct answer.

A **Whip** is a **formal direction issued by a political party to its legislators** to regulate their conduct in the House, particularly to **ensure attendance and voting in line with the party's official position**.

- It is an important instrument for maintaining party discipline in a parliamentary system.
- The term 'whip' is derived from the old British practice of "whipping in" lawmakers to follow the party line.

Statement a is correct: The **office of the Whip is not mentioned anywhere in the Constitution of India or in the Rules of Procedure and Conduct of Business of either House of Parliament**. It is a **parliamentary convention** derived from the British system.

The Leaders and Chief Whips of Recognised Parties and Groups in Parliament (Facilities) Act, 1998 and The Leaders and Chief Whips of Recognised Parties and Groups in Parliament (Telephone and Secretarial Facilities) Rules, 1999 provide for facilities to Leaders and Chief Whips of recognised parties and groups in Parliament.

Statement b is incorrect: The most important role in the whip system is that of the **chief whip** of a political party. There are also **additional whips**, apart from the chief whip.

- In the Indian Parliamentary system, the **Minister of Parliamentary Affairs** (not the Minister of Home Affairs) typically **serves as the Government Chief Whip in the Lok Sabha**.
- In the **Rajya Sabha**, the **Minister of State for Parliamentary Affairs** typically serves as the government's chief whip.

Statement c is correct: Under Paragraph 2(1b) of the **Tenth Schedule (Anti-Defection Law)**, a **Member of Parliament or State Legislature may be disqualified if he or she votes or abstains from voting contrary to any direction (whip) issued by the political party**, without prior permission and without subsequent condonation within the prescribed period. Thus, violation of a valid whip can attract disqualification.

Statement d is correct: The Election Commission of India has cleared that political parties will not be able to issue whips for their Members of Parliament and Members of Legislative Assemblies. This clearly means that the legislatures will not be forced to vote in the poll and can abstain from voting. They will be free to vote or not.

Knowledge Base: **Kinds of Whips in Parliamentary Practice**

In parliamentary practice, whips are generally classified into three categories based on the degree of compulsion involved:

1. One-line whip: This merely informs members about a forthcoming vote or discussion and allows them the discretion to abstain.

2. Two-line whip: This requires members to be present in the House at the time of voting but does not mandatorily direct them on how to vote.

3. Three-line whip: This is the strictest form and directs members to be present and vote strictly in accordance with the party's official position. It is the most commonly issued form of whip in contemporary legislative practice.

Source:) Mastering Indian Polity by Forum IAS, Chapter:16 - Legislative Branch, Page: 363.

<https://indianexpress.com/article/political-pulse/jagdeep-dhankhar-parliament-whip-system-9800118/>

<https://hindi.eci.gov.in/files/file/478-presidential-election-2017-political-parties-cant-issue-whip-says-election-commission/?do=download&r=1028&confirm=1&t=1&csrfKey=bb64bcbf6bc5bc6a16bad2e4a842d065>

<https://www.livemint.com/politics/news/explained-the-crucial-role-of-a-whip-in-assembly-and-parliament-in-5-points-bjp-india-bloc-look-sabha-budget-session-11743503858791.html>

<https://www.thehindu.com/news/national/manish-tewari-introduces-bill-in-lok-sabha-to-free-mps-from-whip-tyranny-promote-good-law-making/article70370886.ece>

<https://www.thehindu.com/news/national/cross-voting-in-rajya-sabha-elections-puts-the-spotlight-back-on-punitive-action-against-rebel-mlas/article67892777.ece>

<https://www.thehindu.com/news/national/tamil-nadu/parties-cannot-issue-whips-to-mlas-in-rajya-sabha-polls/article4824376.ece>

Subject:) Polity

Topic:) Legislature

Subtopic:)

Q.2)

Ans) c

Exp) Option c is the correct answer.

The **Constitution (Forty-second Amendment) Act, 1976** is often called the “**Mini-Constitution**” because it introduced several significant changes in the Indian Constitution, including the addition of new Directive Principles of State Policy (DPSPs). This amendment act **added four new DPSPs into the constitution of India**:

- **Article 39 (f)**: Providing children the opportunities and facilities to develop in a healthy manner (**Option II is correct**).
- Article 39(f) of the Constitution of India states that The State shall, in particular, direct its policy towards securing—that children are given opportunities and facilities to develop in a healthy manner** and in conditions of freedom and dignity and that childhood and youth are protected against exploitation and against moral and material abandonment.
- **Article 39A**: To promote equal justice and to provide free legal aid to the poor.
- **Article 43A**: To take steps to secure the participation of workers in the management of industries (**Option I is correct**).
- **Article 48A**: To protect and improve the environment and to safeguard forests and wildlife (**Option IV is correct**)

Option III is incorrect: To minimize inequalities in income, status, facilities, and opportunities: This is part of **Article 38(2)**, which was **added by the 44th Amendment Act, 1978 (not 1976)**. It did not come from the 42nd Amendment.

Thus, **three of the above DPSPs (I, II, and IV) were added through the 42nd Constitutional Amendment Act, 1976.**

Knowledge Base: DPSPs Added Through Other Amendment Acts:

I. 86th Constitutional Amendment Act, 2002

Modified DPSP:

- **Article 45 – Originally:** Provision for free and compulsory education for children up to 14 years.
- **After the 86th Amendment**, Article 45 was changed to:
- **“Provision for early childhood care and education for children below the age of six years.”**
- This amendment made education for 6–14 years a **Fundamental Right (Article 21A)**, and shifted the DPSP focus toward early childhood care.

II. 97th Constitutional Amendment Act, 2011

Added a new DPSP:

- **Article 43B – Promotion of cooperative societies.**
- The State shall promote the voluntary formation, autonomous functioning, democratic control, and professional management of cooperative societies.

Source:) Mastering Indian Polity by Forum IAS, Chapter: 11 – Directive Principles of State Policy, Page: 175.

https://nios.ac.in/media/documents/SrSec338New/338_Introduction_To_Law_Eng/338_Introduction_To_Law_Eng_L20.pdf Page: 300–301.

<https://egyankosh.ac.in/bitstream/123456789/57875/1/Unit5.pdf> Page: 45–46.

<https://www.mea.gov.in/images/pdf1/part4.pdf>

Subject:) Polity

Topic:) Directive Principles of State Policy

Subtopic:)

Q.3)

Ans) c

Exp) Option c is the correct answer.

Pair I is correct: First Non-Congress Chief Minister in a State – E. M. S. Namboodiripad

E. M. S. Namboodiripad became the Chief Minister of Kerala in 1957 leading a Communist Party of India government. This was the **first time a non-Congress party formed a government in any Indian state after Independence**. It was also the **first democratically elected Communist government in the world**.

Pair II is correct: First Non-Congress Prime Minister of India – Morarji Desai

Morarji Desai became the **Prime Minister in 1977** after the defeat of the Congress in the post-Emergency general elections. He **headed the Janata Party government** and was the **first Prime Minister of India** who **did not belong to the Indian National Congress**.

Pair III is incorrect: Longest Serving Chief Minister - Pawan Kumar Chamling (not Jyoti Basu)

Although **Jyoti Basu** served as **Chief Minister of West Bengal from 1977 to 2000 (over 23 years)**, he is not the longest serving Chief Minister in India. The **record is held by Pawan Kumar Chamling**, who **served as Chief Minister of Sikkim from 1994 to 2019 (over 24 years)**, making him the **longest-serving Chief Minister in Indian history**.

Pair IV is correct: Longest Serving Prime Minister of India - Jawaharlal Nehru

Jawaharlal Nehru served as **Prime Minister** from 15 August 1947 until his death on 27 May 1964, completing **more than 16 years in office**. He remains the **longest-serving Prime Minister of India**.

Narendra Modi is the second longest serving Prime Minister of India serving for nearly 12 years.

Source:) <https://ncert.nic.in/textbook/pdf/leps202.pdf> Page: 33.

<https://ncert.nic.in/textbook/pdf/leps206.pdf> Page: 103.

<https://indianexpress.com/article/india/ten-oaths-nitish-still-not-the-longest-serving-chief-minister-10375821/>

<https://ddnews.gov.in/en/narendra-modi-becomes-indias-second-longest-serving-pm-surpasses-indira-gandhis-record/>

Subject:) Polity

Topic:) Post Independence

Subtopic:)

Q.4)

Ans) b

Exp) Option b is the correct answer.

Article 32 is a **Fundamental Right** enshrined in **Part III** of the Constitution of India that **guarantees the right to move the Supreme Court** for the **enforcement of Fundamental Rights**. It empowers the Supreme Court to issue writs such as Habeas Corpus, Mandamus, Prohibition, Certiorari and Quo Warranto for the protection of these rights.

Dr. B. R. Ambedkar famously **described Article 32 as the “heart and soul” of the Constitution**, stating that the Constitution would be “a nullity” without it.

Statement I is incorrect: Article 32(3) provides that Parliament, by law, may empower any other court to exercise within its local limits all or any of the powers exercisable by the Supreme Court under Article 32. **This power is not vested in the President.**

Statement II is correct: Article 32 is a fundamental right that allows citizens to move the Supreme Court for the enforcement of other fundamental rights.

- The **Supreme Court** has held that a **writ petition under Article 32 can be entertained even in a death penalty case** if there is a clear **violation of procedural safeguards** that amount to **infringement of Fundamental Rights**, particularly under **Articles 14 and 21**.
- Since fairness of procedure is integral to the right to life and equality before law, the Court may exercise its jurisdiction where constitutional infirmities are demonstrated.

Statement III is correct: As per Article 32 of the Constitution of India (Remedies for enforcement of rights conferred by this Part III), the right to move the Supreme Court by appropriate proceedings for the enforcement of the rights conferred by this Part III of the Constitution is guaranteed. Also, the Supreme Court shall have power to issue directions or orders or writs, including writs in the nature of habeas corpus, mandamus, prohibition, quo warranto and certiorari, **whichever may be appropriate, for the enforcement of any of the rights conferred by the Part III of the Constitution of India. Hence Article 32 provides that the writs, directions and orders of the Supreme Court under Article 32 are confined to Fundamental Rights only.**

Also, Article 139 of the Constitution mentions that the Parliament may by law confer on the Supreme Court power to issue directions, orders or writs, including writs in the nature of habeas corpus, mandamus, prohibition, quo warranto and certiorari, or any of them, for any purposes other than those mentioned in clause (2) of article 32. Hence any powers

conferred upon the Supreme Court to issue writs orders etc. for purposes other than the enforcement of Fundamental Rights will be outside the purview of Article 32.

Source:) Mastering Indian Polity by Forum IAS, Chapter: 10 – Fundamental Rights, Page: 153.

<https://indiankanoon.org/doc/981147/>

<https://www.thehindu.com/news/national/andhra-pradesh/article-32-enables-every-citizen-to-approach-supreme-court-for-enforcement-of-fundamental-rights-says-cji/article70287449.ece>

<https://www.thehindu.com/news/national/death-sentence-can-be-challenged-under-article-32-on-ground-of-breach-of-procedural-safeguards-says-supreme-court/article69974305.ece>

Subject:) Polity

Topic:) Fundamental Rights

Subtopic:)

Q.5)

Ans) c

Exp) Option c is the correct answer.

The **Eighth Schedule** of the Constitution of India **lists the officially recognized languages of the Republic of India**. At present, it includes **22 languages**.

Originally, the Constitution recognized 14 languages. The number increased through subsequent Constitutional Amendments:

- 21st Amendment Act, 1967 – Sindhi added
- 71st Amendment Act, 1992 – Konkani, Manipuri, and Nepali added
- 92nd Amendment Act, 2003 – Bodo, Dogri, Maithili, and Santhali added

Option a is incorrect: English, Kashmiri, Punjabi

Kashmiri and Punjabi are included in the Eighth Schedule. English is not included.

Option b is incorrect: Bhojpuri, Gujarati, Urdu

Gujarati and Urdu are included in the Eighth Schedule, but **Bhojpuri is not included** in the Eighth Schedule (though there is a long-standing demand for its inclusion).

Option c is correct: Nepali, Santhali, Dogri – All three are included in the Eighth Schedule.

- **Nepali** was added to the Eighth Schedule by the **71st Constitutional Amendment Act, 1992**.
- **Santhali and Dogri** were added by the **92nd Constitutional Amendment Act, 2003**.
- Hence, this option is correct.

Option d is incorrect: Manipuri, Rajasthani, Bengali

- **Bengali is included** in the Eighth Schedule.
- **Manipuri** was **included** in the Eighth Schedule through the **71st Constitutional Amendment Act, 1992**.
- **Rajasthani is not included**; however, it has a long-standing demand for inclusion.

Therefore, **option a) is the correct answer**.

Knowledge Base: The **languages currently included in the Eighth Schedule are:**

(1) Assamese, (2) Bengali, (3) Gujarati, (4) Hindi, (5) Kannada, (6) Kashmiri, (7) Konkani, (8) Malayalam, (9) Manipuri, (10) Marathi, (11) Nepali, (12) Oriya, (13) Punjabi, (14) Sanskrit, (15) Sindhi, (16) Tamil, (17) Telugu, (18) Urdu (19) Bodo, (20) Santhali, (21) Maithili and (22) Dogri.

Source:)

https://www.mha.gov.in/sites/default/files/EighthSchedule_19052017.pdf

<https://www.thehindu.com/news/national/jmm-bjp-fight-over-inclusion-of-ho-language-in-eighth-schedule/article68651895.ece>

<https://indianexpress.com/article/research/the-eighth-schedule-of-the-indian-constitution-how-language-inclusion-creates-exclusion-9952740/>

Subject:) Polity

Topic:) Language

Subtopic:)

Q.6)

Ans) a

Exp) Option a is the correct answer.

In **Dr. Jaya Thakur v. Union of India case (2026)**, the **Supreme Court** has held that the **right to menstrual health and access to Menstrual Hygiene Management (MHM) measures** in educational institutions **forms an integral part of Article 21** of the Constitution, which guarantees the Right to Life and Personal Liberty.

- The Court observed that the right to life under Article 21 includes the right to live with dignity, bodily autonomy, privacy and health.
- The court held that lack of MHM measures in schools violate the right to privacy and bodily autonomy of students.
- Therefore, ensuring menstrual hygiene facilities in educational institutions is not merely a welfare measure but a constitutional obligation flowing from the expanded interpretation of Article 21.

Source:) <https://www.thehindu.com/news/national/supreme-court-on-menstrual-health-sanitary-pads-toilets-private-govt-schools/article70569703.ece>

<https://www.thehindu.com/opinion/editorial/a-full-stop-on-the-right-to-menstrual-health-and-hygiene/article70583236.ece>

https://api.sci.gov.in/supremecourt/2022/35023/35023_2022_7_1502_68117_Judgement_30-Jan-2026.pdf

Subject:) Polity

Topic:) Fundamental Rights

Subtopic:)

Q.7)

Ans) d

Exp) Option d is the correct answer.

Tribunals are the **quasi-judicial institutions** established to **adjudicate on disputes** pertaining to administration, taxation, environment, securities, etc. They serve as an alternative to the traditional court system and are established to ensure speedy justice, unburden the courts, and provide cost-effective & decentralized resolution of disputes.

The constitutional basis for tribunals is found in **Part XIV-A** of the Constitution, which comprises **Articles 323A and 323B**.

Statement I is incorrect: The **provisions relating to tribunals** were **inserted** into the Constitution **by the 42nd Constitutional Amendment Act, 1976**, not by the 44th Amendment.

The 42nd Amendment inserted **Part XIV-A** (Tribunals) in the Constitution and introduced **Article 323A (Administrative Tribunals)** and **Article 323B (Tribunals for other matters)**, thereby giving constitutional status to tribunals in India.

Statement II is incorrect: Under **Article 323A**, **only Parliament has the power to establish Administrative Tribunals for service matters** of the Union and the States. State Legislatures do not have concurrent authority in this regard.

However, under **Article 323B**, **both Parliament and State Legislatures** can **establish tribunals for specified matters** (such as taxation, labour disputes, land reforms, etc.).

Source:) Mastering Indian Polity by Forum IAS, Chapter:17 - Judicial Branch, Page: 491.

<https://www.thehindu.com/news/national/supreme-court-strikes-down-provisions-of-tribunal-reforms-act-directs-govt-to-establish-national-tribunal-commission/article70297888.ece>

<https://www.thehindu.com/news/national/supreme-court-rules-that-tribunals-cannot-direct-government-to-frame-policy/article67669766.ece>

Subject:) Polity

Topic:) Judiciary

Subtopic:)

Q.8)

Ans) b

Exp) Option b is the correct answer.

Option b is correct: Operation Mahadev was a **joint counter-terrorism mission conducted in July 2025** by the **Indian Army, the Central Reserve Police Force (CRPF), and the Jammu & Kashmir Police.**

- The operation was launched to **track down and eliminate the terrorists responsible for the April 2025 Pahalgam attack**, including its **alleged mastermind**.
- Acting on specific intelligence inputs and supported by aerial surveillance, security forces carried out a coordinated operation in the forested area near Srinagar.
- The mission culminated in the **successful neutralisation of all three militants involved**.

Thus, the **description clearly corresponds to Operation Mahadev**.

Option a is incorrect: Operation Sindoor

Operation Sindoor was a military response launched in May 2025 targeting terrorist infrastructure across the Line of Control in Pakistan and Pakistan-occupied Kashmir after the Pahalgam attack.

It was **not a localised joint counter-terror mission** aimed at eliminating specific terrorists within Jammu & Kashmir.

Option c is incorrect: Operation Shiva

Operation Shiva was initiated by the Indian Army to **ensure comprehensive security arrangements for the annual Amarnath Yatra pilgrimage in Jammu & Kashmir**.

Its objective was pilgrimage protection, not the tracking of specific terrorists linked to the Pahalgam attack.

Option d is incorrect: Operation Keller

This was a **separate counter-insurgency operation in the Keller area of Shopian district**.

Although it involved anti-terror action and **neutralised three Lashkar-e-Taiba (LeT) terrorists**, it was not specifically linked to tracking and eliminating the masterminds of the April 2025 Pahalgam attack.

Source:) <https://www.pib.gov.in/PressReleasePage.aspx?PRID=2149811®=3&lang=2>

<https://indianexpress.com/article/india/amit-shah-terrorists-operation-sindoor-mahadev-pahalgam-attack-parliament-10156610/>

<https://www.newsonair.gov.in/indian-army-launches-operation-shiva-to-ensure-foolproof-security-for-amarnath-yatra-2025/>

<https://www.thehindu.com/news/national/what-is-operation-keller-that-india-launched-after-operation-sindoor/article69574243.ece>

<https://www.pib.gov.in/PressReleasePage.aspx?PRID=2127370>

Subject:) Polity

Topic:) Security

Subtopic:)

Q.9)

Ans) c

Exp) Option c is the correct answer.

The **One Big Beautiful Bill Act of 2025 (H.R. 1)** is a major **federal legislation enacted by the United States Congress** and signed into law by the President of the United States of America in 2025.

- The Act **gained international attention** due to its wide-ranging **provisions relating to federal taxation**, budget reconciliation measures, public expenditure, defense, healthcare, **immigration enforcement** and other significant domestic policy reforms within the U.S. federal framework.

Hence, it is legislation **enacted by the United States of America, not by the European Union, United Kingdom, or Israel**.

Source:)

<https://www.thehindu.com/news/international/one-big-beautiful-bill-passed-whats-in-it-when-it-kicks-in-and-why-it-matters/article69772454.ece>

<https://www.thehindu.com/business/Economy/what-is-president-trumps-one-big-beautiful-bill-all-about-explained/article69773008.ece>

Subject:) International Relations

Topic:) Places in News

Subtopic:)

Q.10)

Ans) a

Exp) Option a is the correct answer.

The **Public Accounts Committee (PAC)** is one of the most important **parliamentary financial committees** of the Parliament of India. It was **first set up in 1921** under the **Government of India Act, 1919**, and continues to function under the parliamentary system established by the Constitution of India.

It acts as a watchdog over the executive to ensure that public money granted by Parliament is spent lawfully and for the intended purposes.

Statement I is correct: This is a primary and core function of the PAC. The PAC examines the annual reports of the **Comptroller and Auditor General of India (CAG) relating to the Appropriation Accounts and Finance Accounts** of the Union Government.

Its primary function is to ensure that expenditure incurred by the executive conforms to the authority granted by Parliament and is legally and properly spent.

Statement II is incorrect: The **Chairperson of the PAC** is appointed by the **Speaker** of the Lok Sabha. Although, since 1967, there has been a strong **parliamentary convention** that the **Chairperson is drawn from the Opposition**, there is no **mandatory provision in the Rules of Procedure** that requires the Chairperson to be from the Opposition party.

Statement III is incorrect: The PAC does not consist exclusively of members from the Lok Sabha. It comprises 22 members (15 from the Lok Sabha and 7 from the Rajya Sabha).

Therefore, members from the Rajya Sabha are also included.

Source:) Mastering Indian Polity by Forum IAS, Chapter:16 - Legislative Branch, Page: 415.

<https://www.thehindu.com/news/national/public-accounts-committee-to-hold-performance-review-of-regulatory-bodies/article68610223.ece>

<https://www.thehindu.com/news/national/pac-questions-fssai-for-unchecked-sale-of-adulterated-food-items/article70308600.ece>

<https://www.thehindu.com/news/national/pac-pulls-up-govt-for-poorly-planned-implementation-of-4455%E2%80%90crore-sankalp-scheme/article70657077.ece>

Subject:) Polity

Topic:) Legislature

Subtopic:)

Q.11)

Ans) d

Exp) Option d is the correct answer.

Statement a is correct: The 1920s were marked by intense debate within the national movement. After the withdrawal of the **Non-Cooperation Movement (1922)**, a strategic question emerged. Should Indians continue the **boycott of councils**, or enter the legislatures and challenge the government from within?

This debate led to the formation of the **Swaraj Party**. However, internal differences soon surfaced. One group favoured strict **obstruction inside councils**. Another supported **conditional cooperation** if meaningful reforms were offered. From this disagreement emerged the **Responsive Co-operation Party**. It represented a strand of nationalist politics committed to **constitutional methods** and **measured engagement** with colonial authority.

Statement b is correct: **Responsive Co-operation Party** was formally organised at the **Akola Conference (1925) in Maharashtra**. Leaders like **N. C. Kelkar** and **B. S. Moonje** articulated the idea of “responsive cooperation.” They believed that politics required **flexibility**, not rigid rejection of every reform.

Statement c is correct: The term “Responsive” was central to its ideology. It did not mean blind support. It meant cooperation only if the government showed **genuine responsiveness** to Indian demands. The party viewed **legislative councils as tactical platforms** for negotiation and gradual reform. Council entry was seen as a **political instrument**, not a betrayal of nationalism. Their larger objective remained **self-government through constitutional politics**.

Statement d is incorrect: Importantly, the Responsive Co-operation Party was **not the official parliamentary wing of the Congress**. It was a **breakaway faction** arising from disagreements within the Swaraj Party. It functioned independently and was never formally adopted by the Congress as its authorised legislative arm.

Source:) <https://www.indianculture.gov.in/digital-district-repository/district-repository/responsive-cooperative-party>

Subject:) History

Topic:) Non-Cooperation Movement

Subtopic:)

Q.12)

Ans) c

Exp) Option c is the correct answer.

The **Government of India Act, 1919** introduced major constitutional changes in provincial administration. It attempted to widen Indian participation while retaining decisive British control. The most striking innovation was the introduction of **Dyarchy in provinces**, which divided subjects into ‘Reserved’ and ‘Transferred’ categories. However, despite this apparent devolution, ultimate authority continued to remain firmly in British hands.

Statement I is correct. Under the Act, Governors were provided with an **Instrument of Instructions**. This document laid down the principles they were expected to follow in governance. It defined their special responsibilities and discretionary powers. It acted as a constitutional guide and strengthened their authority within the dyarchy system.

Statement II is correct. The reserved subjects, such as police, justice and revenue, were administered by the Governor and his Executive Council. These officials were appointed by the British Government. They were not responsible to the provincial legislature. Instead, they were jointly responsible to the **Governor-General and the Secretary of State for India**. This arrangement clearly shows that real authority continued to flow upward within the imperial structure.

Source:) <https://egyankosh.ac.in/bitstream/123456789/89553/3/Unit-10.pdf> – Unit 10 – Constitutional Reforms – Changes in the Provincial Government – Page 141; <https://egyankosh.ac.in/bitstream/123456789/19084/1/Unit-3.pdf> – Unit 3 – Reforms in British Administration: 1858 to 1919 – The Montague-Chelmsford Reforms 1919 – Page 44

Subject:) History

Topic:) Governance during British rule

Subtopic:)

Q.13)

Ans) a

Exp) Option a is the correct answer.

Between the **6th and 11th centuries CE**, several mahakavyas were composed drawing inspiration from the Mahabharata. Among them, three stand out prominently which are **Kirtarjuniya, Shishupalavadha, and Naishadhiyacharita**.

Shishupalavadha occupies a special place in this tradition. It blends **heroic narrative, poetic ornamentation and philosophical reflection** in a manner that enriches classical Sanskrit literature.

Shishupalavadha was composed by **Magha**, one of the most celebrated Sanskrit poets after Bharavi. He is generally placed around the **7th–8th century CE**.

His epic narrates the episode of **Krishna slaying Shishupala**, based primarily on the Sabha Parva of the Mahabharata and also echoed in the Bhagavata Purana. The poem consists of **twenty cantos and around 1550 verses**.

Magha is admired for his **linguistic mastery, rich similes and intellectual depth**. His poetry combines scholarship with aesthetic charm. In this work, Krishna is portrayed not merely as a divine figure but as an **ideal hero embodying bravery, patience and dignity**.

Knowledge Base:

- **Bharavi** was the author of the celebrated epic Kiratarjuniya. He is generally placed in the **6th century CE**. His work focuses on the episode of **Arjuna's penance and his encounter with Lord Shiva in the guise of a hunter (Kirata)**.
- **Bhasa** was an earlier Sanskrit dramatist. He is known for plays such as Swapnavasavadatta and Urubhanga
- **Shudraka** is traditionally credited with the Sanskrit play Mrichchhakatika (The Little Clay Cart). He is associated with **social drama and urban life themes**, rather than epic poetry.

Source:) <https://egyankosh.ac.in/bitstream/123456789/96151/1/Unit-11.pdf> - Unit 11 - Indian Classical Literature: An Introduction- Sanskrit Epics after Kalidas - Page 175

Subject:) Art and Culture

Topic:) Literature

Subtopic:)

Q.14)

Ans) c

Exp) Option c is the correct answer.

Recently, the **Central Council for Research in Ayurvedic Sciences (CCRAS)** under the Ministry of AYUSH revived two rare manuscripts, Dravyaratnakara Nighantu and Dravyanamakara Nighantu. Their revival is significant because they belong to the rich tradition of **classical Ayurvedic literature**, especially the branch dealing with medicinal substances. These works are not ritual manuals or astronomical treatises. They are specialised **Ayurvedic lexicons (Nighantus)** that document medicinal drugs, their synonyms, properties and therapeutic uses.

Dravyaratnakara Nighantu was authored by **Mudgala Pandita in 1480 AD**. It is an important **Ayurvedic lexicon** consisting of **eighteen chapters**. The text systematically explains **drug synonyms, medicinal properties, and therapeutic actions** of various substances used in Ayurveda. This work draws inspiration from earlier classical compilations such as the **Dhanvantari Nighantu** and the **Raja Nighantu**, yet it also introduces several new medicinal substances. These include materials derived from **plant, mineral, and animal origins**, showing the wide scope of Ayurvedic pharmacology.

The text holds special importance in the branch of **Dravyaguna**, which studies the **pharmacological qualities and therapeutic applications of medicinal substances** in Ayurveda. Because of its detailed treatment of drugs and their properties, it serves as a foundational reference for students and practitioners of classical Ayurvedic medicine.

Dravyanamakara Nighantu is attributed to **Bhisma Vaidya**. Unlike general lexicons, this text serves as a **specialised appendix to the Dhanvantari Nighantu**. Its primary focus is on **homonyms of drug and plant names**, which means it carefully clarifies substances that share similar names but differ in identity. This distinction is extremely important in Ayurvedic practice, because correct identification of medicinal substances is essential for safe and effective treatment. The text contains **182 verses**, along with colophon verses, and has been regarded as a valuable tool for avoiding confusion in pharmacological application.

Such works are particularly significant for disciplines like **Rasashastra** and **Bhaishajya Kalpana**, which deal with the **processing, formulation, and preparation of Ayurvedic medicines**. By ensuring clarity in terminology and substance identification, the text strengthens the practical foundations of Ayurvedic therapeutics.

Knowledge Base:

- **Astronomy** in ancient India is associated with texts like **Surya Siddhanta** and works of scholars such as Aryabhata and Varahamihira.
- Texts dealing with **Vedic rituals** are usually found in the **Brahmanas, Kalpa Sutras and Grihya Sutras**
- Ancient Indian **mathematical literature** includes works such as **Aryabhatiya and Lilavati**. These deal with arithmetic, algebra and geometry.

Source:) <https://government.economictimes.indiatimes.com/news/healthcare/ccras-revives-2-rare-ayurvedic-manuscripts-bridging-ancient-wisdom-with-modern-research/120985455> ;

<https://www.pib.gov.in/PressReleasePage.aspx?PRID=2127487®=3&lang=2>

Subject:) Art and Culture

Topic:) Literature

Subtopic:)

Q.15)

Ans) b

Exp) Option b is the correct answer.

The UNESCO Creative Cities Network (UCCN) recognises cities that place **creativity and cultural heritage at the centre of sustainable urban development**. One of its categories is **Gastronomy**, which honours cities with rich culinary traditions, strong food culture, and active efforts to preserve and promote their gastronomic heritage. Recognition is not given merely for having famous dishes. It is granted when a city demonstrates:

- Deep historical culinary roots
- Unique food traditions
- Community involvement in preserving food heritage
- Integration of gastronomy into economic and cultural planning

Option I is incorrect. Jaipur is recognised under the category of **Crafts and Folk Arts**. The city is known for its **handicrafts, blue pottery, block printing, and traditional arts**, not for gastronomy under UNESCO classification.

Option II is correct. Hyderabad was designated as a **City of Gastronomy in 2019**. The recognition was given because of:

- Its **centuries-old culinary tradition** influenced by Nizami and Mughal culture
- Iconic dishes such as **Hyderabadi Biryani, Haleem, and Kebabs**
- Strong integration of food culture into urban identity and tourism
- Preservation of traditional cooking practices

Hyderabad became **India's first City of Gastronomy** under UCCN.

Option III is correct. Lucknow was recently inducted as a **City of Gastronomy**. The recognition highlights:

- The rich **Awadhi culinary tradition**
- Heritage dishes like **Tunday Kebabs, Galouti Kebabs, Nihari, and Awadhi Biryani**
- The legacy of **royal Nawabi cuisine**
- Ongoing efforts to preserve culinary craftsmanship

Lucknow's food culture reflects refinement, slow cooking techniques such as **Dum Pukht**, and strong cultural continuity.

Option IV is incorrect. Delhi has not been recognised as a City of Gastronomy under UNESCO. While Delhi has diverse food traditions, it has not received formal designation in this category.

Source:) <https://www.thehindu.com/news/national/uttar-pradesh/lucknow-enters-unesco-list-as-creative-city-of-gastronomy/article70229854.ece> ; https://www.unesco.org/en/creative-cities/grid?fq%5Bsm_unsc_hierarchical_field_ref_datasets_filters_label%5D%5BCreative+field%5D%5B%5D=Creative+field+%7C-%3E+Gastronomy&date_from=&date_to=&query=&hub=80094&op=Search#toggle-facets

Subject:) Art and Culture

Topic:) Miscellaneous

Subtopic:)

Q.16)

Ans) c

Exp) Option c is the correct answer.

The **Rajya Sabha** is the **Upper House of the Parliament** of India. It represents the States and Union Territories in the federal structure and is therefore known as the **Council of States**.

It is a **permanent body** and is **not subject to dissolution**; however, **one-third of its members retire every two years**. The **Vice-President of India** acts as the **ex-officio Chairman** of the Rajya Sabha.

The Rajya Sabha plays a crucial role in maintaining the federal balance and **possesses certain exclusive constitutional powers**, particularly under **Articles 249, 312, and 67(b)**, which distinguish it from the Lok Sabha.

Option I is correct: Under **Article 249**, the **Rajya Sabha** may pass a **resolution** (by a two-thirds majority of members present and voting) declaring that it is **necessary in the national interest for Parliament to legislate on a matter in the State List**.

This power is exclusive to the Rajya Sabha, reflecting its federal character as the Council of States.

Option II is correct: Under **Article 312**, the **Rajya Sabha** can pass a **resolution** (by a two-thirds majority of members present and voting) declaring it necessary or expedient in the national interest to **create new All-India Services**. Lok Sabha cannot initiate this authorisation. Hence, **this power is also exclusive to the Rajya Sabha**.

Option III is correct: **This is an exclusive power of the Rajya Sabha** (under Article 67b)). A **resolution for removal of the Vice-President** (who is ex-officio Chairman of Rajya Sabha) **can be initiated only in Rajya Sabha** (by effective majority, i.e., majority of all then members of Rajya Sabha).

It must then be agreed to by the Lok Sabha (simple majority). **Lok Sabha cannot initiate the motion**.

Option IV is incorrect: Under **Article 248** and Entry 97 of the Union List, **residuary powers** (subjects not enumerated in Union List, State List, or Concurrent List) are **vested in Parliament**. The Rajya Sabha has no special or exclusive role in determining what constitutes residuary subjects.

Source:) Mastering Indian Polity by Forum IAS, Chapter:16 - Legislative Branch, Page: 401.

<https://egyankosh.ac.in/bitstream/123456789/57908/1/Unit7.pdf> Page: 65

Subject:) Polity

Topic:) Legislature

Subtopic:)

Q.17)

Ans) a

Exp) Option a is the correct answer.

The **euro area**, also known as the **eurozone**, refers to the **group of member states of the European Union (EU)** that have **adopted the euro (€) as their official currency**.

- It is a monetary union established to promote deeper economic integration, price stability, and ease of trade among participating countries.
- Monetary policy for the euro area is managed by the **European Central Bank**, headquartered in **Frankfurt, Germany**.
- Currently, **21 out of 27 EU member states form part of the euro area**.

Option I is correct: Bulgaria

Bulgaria adopted the euro on 1 January 2026, thereby becoming the **21st member of the euro area**.

Option II is incorrect: Denmark

Denmark is an EU member but has a formal opt-out from adopting the euro under the Maastricht Treaty. It continues to use the Danish krone and is **not part of the euro area**.

Option III is correct: Austria

Austria is a full member of the euro area and uses the euro as its official currency.

Option IV is incorrect: Sweden

Sweden is an EU member but has not adopted the euro and continues to use the Swedish krona. **It is not part of the eurozone**.

Option V is incorrect: Poland

Poland is also an EU member but has **not adopted the euro** and retains the Polish złoty.

Thus, **only Bulgaria and Austria** are **members of the euro area among the countries listed**.

Therefore, the **correct answer** is: a) **Only two**.

Knowledge Base: Members of the Euro Area (21 Countries)

The 21 countries that currently constitute the euro area are: Austria, Belgium, Bulgaria, Croatia, Cyprus, Estonia, Finland, France, Germany, Greece, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, the Netherlands, Portugal, Slovakia, Slovenia and Spain.

Source:) https://european-union.europa.eu/institutions-law-budget/euro/countries-using-euro_en
<https://www.thehindu.com/news/international/bulgaria-becomes-21st-member-to-adopt-euro-after-eu-green-light/article69789136.ece>
<https://indianexpress.com/article/explained/explained-economics/bulgaria-eurozone-member-russia-10450557/>
 Subject:) International Relations
 Topic:) Places in News
 Subtopic:)

Q.18)

Ans) b

Exp) Option b is the correct answer.

Statement I is incorrect: The Constitution (Article 200) does not prescribe any specific time-limit within which the Governor must act on a Bill.

Just like the President under Article 111, the Governor can delay assent, which gives rise to the concept of a pocket veto (though the term is not mentioned in the Constitution).

Statement II is correct: If a Bill endangers or derogates from the position/powers of the High Court, the Governor is constitutionally bound to reserve it for the consideration of the President.

This is a mandatory reservation under Article 200.

Statement III is correct: The Governor cannot return a Money Bill to the State Legislature for reconsideration. The Governor can either give assent or withhold assent or may reserve it for the consideration of the President.

Knowledge base: When a Bill is reserved by Governor for consideration of the President:

- Under Article 201, the President may either give assent or withhold assent.
- However, in the case of a **non-Money Bill**, the President may direct the Governor to return the Bill to the State Legislature for reconsideration.

Source:) Mastering Indian Polity by Forum IAS, Chapter:15 - Executive Branch, Page: 307 & 316.

<https://www.thehindu.com/news/national/governors-cannot-sit-over-bills-endlessly-say-supreme-court-judges-presidential-reference/article70004889.ece>

<https://www.thehindu.com/opinion/editorial/blow-to-states-on-the-supreme-court-and-state-bills/article70308688.ece>

Subject:) Polity

Topic:) Exercise

Subtopic:)

Q.19)

Ans) b

Exp) Option b is the correct answer.

Information in row I is correct: YUDH ABHYAS

Yudh Abhyas is an annual bilateral Army exercise between India and the United States.

- The 21st edition (2025) was conducted in September 2025 at Fort Wainwright, Alaska (USA).
 - The exercise focuses on high-altitude warfare, counter-terrorism operations, interoperability, and joint tactical drills.
- All three components; exercise name, venue (Alaska, USA), and partner country (USA) are correctly matched.

Information in row II is incorrect: Desert Cyclone-II

Desert Cyclone is a bilateral Army exercise between India and the United Arab Emirates (UAE), not Israel.

- The 2nd edition (Desert Cyclone-II) was held in December 2025 in Abu Dhabi, UAE.
- It focuses on desert warfare tactics and counter-terrorism operations in semi-urban terrain.

Thus, both the location (Rajasthan) and the partner country (Israel) are incorrect.

Information in row III is incorrect: NOMADIC ELEPHANT

Nomadic Elephant is a bilateral Army exercise between India and Mongolia.

- The 2025 edition was conducted in June 2025 at Ulaanbaatar, Mongolia.
- The exercise emphasizes counter-insurgency operations in mountainous terrain under UN peacekeeping scenarios. Although the partner country (Mongolia) is correct, the location (Himachal Pradesh, India) is incorrect.

Information in row IV is correct: Exercise SHAKTI

Exercise Shakti is a bilateral Army exercise between India and France, conducted alternately in both countries.

- The 2025 edition (8th edition) was conducted in June 2025 at La Cavalerie in France.
- The aim of the exercise is to enhance the joint military capability of both sides to undertake multi-domain operations in a sub-conventional scenario.

All three components; exercise name, venue (La Cavalerie, France), and partner country (France) are correctly matched.

Therefore, **only two rows are correctly matched.**

Hence, the **correct answer is: a) Only two.**

Source:) <https://www.pib.gov.in/PressReleasePage.aspx?PRID=2162738>

<https://www.pib.gov.in/PressReleasePage.aspx?PRID=2205132®=3&lang=2>

<https://www.pib.gov.in/PressReleasePage.aspx?PRID=2132473>

<https://www.newsonair.gov.in/india-france-joint-military-drill-exercise-shakti-2025-to-be-held-in-france-from-june-18-to-july-1/>

<https://www.newsonair.gov.in/india-uae-desert-cyclone-ii-exercise-concludes-in-abu-dhabi/>

Subject:) International Relations

Topic:) Defence Exercises

Subtopic:)

Q.20)

Ans) b

Exp) Option b is the correct answer.

The **Union Executive** under the Constitution of India refers to the branch of government responsible for the administration and execution of laws at the Union level. It is dealt with in **Chapter I, Part V** of the Constitution of India (**Articles 52–78**).

The Union Executive comprises the **President, Vice-President, Prime Minister, Council of Ministers, and the Attorney General of India**.

The President is the constitutional head of the Union, while the real executive authority is exercised by the Council of Ministers headed by the Prime Minister under Article 74.

Option I is correct: Prime Minister

The **Prime Minister is part of the Union Executive under Article 74**, which provides for a Council of Ministers headed by the Prime Minister to aid and advise the President. The Prime Minister of India is the **real executive authority in India**.

Option II is correct: President

The President of India (**Article 52**) is the **formal executive head of the Union**. All executive actions of the Government of India are taken in his/her name (Article 53).

Option III is correct: Vice-President

The Vice-President of India is provided under **Article 63** and forms part of the constitutional scheme of the Union Executive. Though primarily the ex-officio Chairman of the Rajya Sabha, the **office forms part of the Union Executive under the Constitution**.

Option IV is incorrect: Solicitor General of India

This is **not a constitutional office**. It is a statutory post (second-highest law officer, subordinate to the Attorney General). The Constitution does not mention it, so **it is not part of the Union Executive as per the Constitution**.

Option V is correct: Attorney General of India

The Attorney General for India (**Article 76**) is the **highest law officer of the Union**. He/she is the chief legal advisor to the Government of India and is a constitutional office forming **part of the Union Executive under the Constitution of India**.

Option VI is incorrect: Comptroller and Auditor General of India

The Comptroller and Auditor General of India is a **constitutional authority under Article 148**. However, the CAG is an independent authority responsible for auditing government accounts and ensuring financial accountability. The office is placed in Part V, Chapter V, not under the Union Executive chapter. Therefore, **it is not part of the Union Executive**.

Source:) Mastering Indian Polity by Forum IAS, Chapter:15 - Executive Branch, Page: 260

<https://knowindia.india.gov.in/profile/the-union/executive.php>

<https://www.legislative.gov.in/static/uploads/2025/07/ca7ce5c746fa7480804bbdeb6cb704f0.pdf>

Subject:) Polity

Topic:) Executive

Subtopic:)

Q.21)

Ans) b

Exp) Option b is the correct answer.

The **Third Schedule of the Constitution of India** contains the prescribed **Forms of Oaths or Affirmations** for various constitutional functionaries. These forms are linked to specific Articles such as Articles 75, 99, 124(6), 148(2), 164, 188 and 219.

Option I is incorrect. The oath of the **President of India** is **not** provided in the Third Schedule. It is prescribed separately under **Article 60** of the Constitution.

Option II is incorrect. The oath of the **Vice-President of India** is not provided in the Third Schedule. It is prescribed separately under **Article 69** of the Constitution.

Option III is correct. The Form of oath or affirmation to be made by a candidate for election to Parliament is included under Third Schedule of the Constitution of India.

Option IV is correct. The oath of the Comptroller and Auditor-General of India is prescribed in the Third Schedule and is read with **Article 148(2)**.

Knowledge Base: **Forms of Oaths or Affirmations under the Third Schedule:**

S. No.	Authority / Office Holder	Relevant Article	Form of Oath/Affirmation (Third Schedule)
1	The Union Ministers	Article 75(4)	Oath of Office and Oath of Secrecy
2	A Candidates for election to Parliament	Article 84(a)	Oath/Affirmation before authorized person
3	Members of Parliament	Article 99	Oath/Affirmation before taking seat
4	Judges of the Supreme Court	Article 124(6)	Oath/Affirmation before the President or appointed person
5	Comptroller and Auditor-General of India	Article 148(2)	Oath/Affirmation before the President
6	State Ministers	Article 164(3)	Oath of Office and Oath of Secrecy
7	Candidates for election to State Legislature	Article 173(a)	Oath/Affirmation before authorized person
8	Members of State Legislature	Article 188	Oath/Affirmation before taking seat
9	Judges of the High Courts	Article 219	Oath/Affirmation before the Governor or appointed person

Source:) <https://www.mea.gov.in/Images/pdf1/S3.pdf> ;

<https://www.legislative.gov.in/static/uploads/2025/07/ca7ce5c746fa7480804bbdeb6cb704f0.pdf>

Subject:) Polity

Topic:) Schedules

Subtopic:)

Q.22)

Ans) a

Exp) Option a is the correct answer.

The Population Census is not a routine survey. It is the **largest administrative exercise undertaken by the Government of India**, forming the base for policy planning, welfare targeting, delimitation, and fiscal transfers. Its conduct is rooted in clear constitutional placement and statutory backing. The upcoming Census is significant because it introduces structural changes in methodology and scope.

Statement I is correct. Census is placed in the Union List of the Seventh Schedule, which means only Parliament has the authority to legislate on it. The process is governed by the **Census Act, 1948**, which empowers the Central Government to notify the Census, appoint the Census Commissioner, and supervise the entire operation. This centralised framework ensures **uniform standards across all States and Union Territories**, which is essential for producing comparable national data.

Statement II is incorrect. Census 2027 will be the 16th Census in the country and 8th after independence. Census is the biggest source of primary data at village, town and ward level providing micro level data on various parameters including Housing Condition; Amenities & Assets, Demography, Religion, SC & ST, Language, Literacy & Education, Economic Activity, Migration and Fertility. The Census Act, 1948 and the Census Rules, 1990 provide the legal framework for conduct of Census.

Statement III is correct. The Government has officially announced that the upcoming Census will include **enumeration of castes for the first time in independent India**. The last census where caste was enumerated for Hindus was in 1931. After Independence, the Census has collected caste data only for **Scheduled Castes and Scheduled Tribes**. A comprehensive caste enumeration has not been part of the regular Census exercise. Therefore, the inclusion of caste data in Census 2027 marks a major policy shift with administrative and socio-economic implications.

Statement IV is correct. The upcoming Census has been described as **India's first digital Census**. Until 2011, data collection was paper-based. The move to digital mode represents a structural transformation in how enumeration will be conducted, stored, and processed. It is expected to improve **speed, accuracy, and data validation**, and reduce delays between enumeration and release of results.

- Use of mobile apps for data collection and the Central portal for monitoring purposes will ensure better quality data.
- Data dissemination will be much better and in a user friendly way so that all the queries on required parameters for policy making will be made available on the click of a button.
- Census-as-a-service (CaaS) will deliver data to ministries in a clean, machine-readable and actionable format.

Source:) <https://www.thehindu.com/news/national/what-is-the-significance-of-the-census-explained/article69672320.ece> ; <https://www.thehindu.com/news/national/centre-announces-population-census-caste-to-be-counted/article69656894.ece> ; <https://www.thehindu.com/news/national/census-2027-to-retain-2011-census-definition-of-an-urban-area/article69969030.ece> ;

<https://www.pib.gov.in/PressReleasePage.aspx?PRID=2202983®=3&lang=1>

Subject:) Polity

Topic:) Miscellaneous

Subtopic:)

Q.23)

Ans) c

Exp) Option c is the correct answer.

The **Right to Equality (Articles 14–18)** forms the first group of **Fundamental Rights** under **Part III** in the Constitution. It is the constitutional guarantee that the State will treat individuals fairly in law and in public life. It addresses discrimination, social hierarchy and unequal access to public opportunities

Option a is incorrect. Abolition of untouchability (Article 17) is expressly included under the Right to Equality. This provision directly attacks social exclusion rooted in caste practices. It is firmly placed within the Right to Equality because it removes a structural barrier to equal status in society.

Option b is incorrect. Equality of opportunity in matters of public employment (Article 16) is a core component of the Right to Equality. It ensures that access to government jobs is not denied on arbitrary or discriminatory grounds. Since public employment is a key avenue of State participation, this provision strengthens substantive equality.

Option c is correct. Prohibition of traffic in human beings and forced labour (Article 23) falls under the **Right against Exploitation (Articles 23–24)**. This right protects individuals from economic and physical exploitation, not from inequality in status or legal treatment.

Option d is incorrect. Abolition of titles (Article 18), except military and academic distinctions, is also included under the Right to Equality. The objective here is to prevent artificial social ranking and privilege. By removing hereditary or honorific titles, the Constitution reinforces the idea of equal civic status.

Source:) The Constitution of India – Part III – Fundamental Rights;

Mastering Indian Polity by Forum IAS - 0th Edition – Chapter 10 – Fundamental Rights – Classification of Fundamental Rights – Page 106;

Subject:) Polity

Topic:) Fundamental Rights

Subtopic:)

Q.24)

Ans) c

Exp) Option c is the correct answer.

Part XIII of the Constitution of India safeguards the economic unity of India by ensuring freedom of trade, commerce and intercourse across the country. **Article 304** forms an important limitation clause. It allows States to impose certain taxes and restrictions, but strictly prohibits discrimination against goods from other States. The recent Supreme Court decision striking down a discriminatory VAT notification has reaffirmed this constitutional discipline.

Statement I is correct. Article 304 (a) allows a State to tax goods imported from other States. However, the tax must be equal to that imposed on similar goods produced within the State. The Constitution clearly prohibits discriminatory taxation. **The Supreme Court recently reiterated that taxation cannot be structured to give local goods a price advantage over similar goods coming from outside the State.**

Statement II is correct. Although **Article 302** empowers Parliament to impose restrictions in public interest, **Article 303** prohibits Parliament from giving preference to one State over another or discriminating between States. **Exceptions exist in certain cases of scarcity of goods in any part of the territory of India.**

Article 303 (b) of the Constitution of India states that Parliament can make any law giving, or authorising the giving of, any preference to one state over another or making, or authorising the making of, any discrimination between states if it is declared by such law that it is necessary to do so for the purpose of dealing with a situation arising from scarcity of goods in any part of the territory of India.

Source:) The Constitution of India – Part XIII – Trade, Commerce And Intercourse Within The Territory of India – Article 301 to 307;

<https://indianexpress.com/article/legal-news/taxation-cannot-be-weaponised-to-discriminate-against-goods-imported-from-other-states-supreme-court-10270649/>

Subject:) Polity

Topic:) Fundamental Rights

Subtopic:)

Q.25)

Ans) d

Exp) Option d is the correct answer.

Article 75 of the Constitution lays down the key provisions relating to the appointment, tenure and responsibility of the Union Council of Ministers. Important changes were introduced by the **91st Constitutional Amendment Act, 2003**, particularly regarding the size of the Council and other issues.

Statement I is incorrect. **Article 75(1A)**, inserted by the 91st Constitutional Amendment Act, provides that the total number of Ministers, including the Prime Minister, **shall not exceed fifteen per cent of the total strength of the House of People (its the total strength of the House of People not the then members of the House of People because the then members mean the total number of members minus the vacancies)**. This provision was introduced to curb the practice of excessively large ministries and to ensure better governance and fiscal responsibility.

Statement II is incorrect. The Constitution does **not prescribe any minimum number** of Ministers at the Union level. There is only a maximum limit of fifteen per cent of the Lok Sabha strength. Unlike some State-level provisions, no constitutional clause mandates a minimum strength for the Union Council of Ministers.

Statement III is incorrect. Under **Article 75(5)**, a person who is not a member of either House of Parliament may still be appointed as a Minister. However, a Minister who for any period of six consecutive months is not a member of either House of Parliament shall at the expiration of that period cease to be a Minister.

Thus, membership of Parliament is not a pre-condition for taking oath, but it is mandatory for continuation as a minister beyond six months.

Source:) The Constitution of India - Part V - The Union - Council of Ministers - Article 75; Mastering Indian Polity by Forum IAS - 0th Edition - Chapter 15 - Executive Branch - Other provisions related to the Council of Ministers - Article 75 - Page 291

Subject:) Polity

Topic:) Executive

Subtopic:)

Q.26)

Ans) c

Exp) Option c is the correct answer.

Under the **Ministry of Education's, Understanding Lifelong Learning for All in Society (ULLAS) - Nav Bharat Saaksharta Karyakram**, the Government of India aims to achieve functional literacy among non-literate persons aged 15 years and above. Under the programme, **any State or Union Territory with a literacy rate exceeding 95% is deemed 'fully literate'**. This programme goes beyond basic reading and writing and includes **foundational literacy and numeracy, digital literacy, financial awareness and civic understanding**.

On **June 24, 2024**, **Ladakh** was declared the **first fully literate Union Territory** of India. This milestone was achieved through large-scale mobilisation of volunteers, local administration and community participation under ULLAS.

Among States, **Tripura, Mizoram, Goa and Himachal Pradesh** have been declared fully literate in recent years. Himachal Pradesh became the latest State to achieve this status in 2025. These achievements reflect the expansion of literacy from a narrow definition to a broader concept that includes digital and life skills.

Source:) <https://www.thehindu.com/news/national/himachal-pradesh/himachals-education-push-literacy-rate-over-99-fewer-than-57000-left-illiterate/article70056697.ece> ;

<https://www.pib.gov.in/PressReleasePage.aspx?PRID=2164749®=3&lang=2>

<https://ullas.education.gov.in/nlp/about-us>

Subject:) Polity

Topic:) Miscellaneous

Subtopic:)

Q.27)

Ans) c

Exp) Option c is the correct answer.

Pair I is correct. Venezuela has been prominently in the news following confirmed actions taken by the United States against its President, Nicolas Maduro. The development attracted global attention because it involved direct intervention by a major power in the political leadership of another sovereign country and causing political instability in the country. Such an event has implications for international law, sovereignty and diplomatic relations as well.

Pair II is correct. Senegal witnessed a significant geopolitical development when France in July 2025, formally handed over its last military base in the country. This marked the end of France's permanent troop presence there, closing a chapter of long-standing post-colonial military engagement in West Africa.

Pair III is correct. The President of Peru Dina Boluarte was removed from office by Peru's Congress on October 10, 2025, after an impeachment process in which lawmakers said she had been unable to tackle the crime wave in the country. She was immediately replaced by Jose Jeri, the legislative body's leader.

Peru's new President Jose Jeri declared a state of emergency in Lima, the capital city of Peru, on October 21, 2025 in the government's latest attempt to stem a surge in violence that has stirred protests and contributed to the downfall of his predecessor.

Source:) <https://indianexpress.com/article/world/us-attack-venezuela-capture-president-nicolas-maduro-10453399/> ; <https://www.thehindu.com/news/international/french-army-leaves-senegal-ending-military-presence-in-west-africa/article69822997.ece> ; <https://www.thehindu.com/news/international/perus-president-jose-jeri-declares-state-of-emergency-in-lima-to-tackle-surge-violence/article70188914.ece>

Subject:) International Relations

Topic:) Places in News

Subtopic:)

Q.28)

Ans) a

Exp) Option a is the correct answer.

Greenland, though sparsely populated, occupies a critical position in the northern hemisphere. The United States' interest in Greenland has to be examined from a strategic and security perspective, particularly in the context of Arctic militarisation and emerging global power competition.

Statement I is correct. The United States views Greenland officially through the lens of national security. Its location in the Arctic enables monitoring of trans-Arctic air and sea routes, including potential missile trajectories between Eurasia and North America. In strategic terms, Greenland acts as a forward observation and defence point in the northern hemisphere. Therefore, security considerations form the primary basis of U.S. interest.

Statement II is correct and explains statement I. Greenland is strategically positioned between the U.S., Canada, the EU and Russia. So, anyone who controls Greenland controls the Arctic region. This has made the region even more attractive to the U.S.

Moreover, due to the melting of the Arctic, the ocean around Greenland is also changing. Many economies view this as an opportunity for opening new sea routes. The number of ships passing through the Arctic has increased by 30%. Compared to the route through the Suez Canal, the journey through the northern route via the Arctic can save weeks of travel time and fuel costs. And Greenland is at the heart of this route.

Source:) <https://www.thehindu.com/news/international/why-is-the-us-targeting-greenland-explained/article70495700.ece>; <https://www.thehindu.com/news/international/what-lies-behind-trumps-greenland-moves-explained/article70547395.ece>

Subject:) International Relations

Topic:) Places in News

Subtopic:)

Q.29)

Ans) a

Exp) Option a is the correct answer.

Ocean waves are generated by wind when it blows over a sizable area (known as fetch) of the ocean. A large fetch and powerful winds that prevail for a longer duration generate higher sea waves. When directly generated and affected by local winds, the school of the waves is called a wind sea, and it is composed of many uneven waves. **After the wind ceases to blow, and the wind waves have left the fetch, their wavelength increases, while their roughness is reduced. These waves are called swell. Swell waves comprise wind-generated waves that are not significantly affected by the local wind.**

Statement I and II are correct and statement II explains statement I: Swell waves are generated by strong and persistent wind systems, often far from the coast. They are characterised by **long wavelengths and long wave periods**, which enable them to travel thousands of kilometres with minimal energy loss. Because of this property, swell waves can strike distant coastlines with considerable force even days after their formation.

Source: <https://egyankosh.ac.in/bitstream/123456789/79951/1/Unit-10.pdf>

<https://www.thehindu.com/sci-tech/energy-and-environment/what-are-swell-waves-explained/article68755965.ece>

Subject:) Geography

Topic:) Oceanography

Subtopic:)

Q.30)

Ans) d

Exp) Option d is the correct answer.

Sustainable Aviation Fuel (SAF) has become central to global efforts to reduce emissions from the aviation sector. It is viewed as a practical pathway for decarbonisation because it can be integrated into existing fuel systems. Accordingly, International and national targets have been announced to promote its gradual adoption.

Option a is correct. SAF is produced from **non-fossil sources** such as used cooking oil, plant waste and agricultural biomass. Unlike conventional aviation turbine fuel, which is derived from crude oil, SAF aims to reduce lifecycle greenhouse gas emissions.

Option b is correct. SAF is compatible with existing aircraft engines when blended with conventional jet fuel. This **drop-in capability** allows airlines to use SAF without modifying current aircraft or refuelling infrastructure.

Option c is correct. India has announced a phased blending roadmap under ICAO's **Carbon Offsetting and Reduction Scheme for International Aviation (CORSIA)**, starting with a 1% blend by 2027, increasing to 2% by 2028, and reaching **5% SAF blending target by 2030** for international flights. This aligns with global efforts to gradually scale up sustainable aviation fuel usage.

Option d is incorrect. Currently, **challenges such as high production costs, limited feedstock availability, and infrastructure compatibility hinder widespread adoption of Sustainable Aviation Fuel (SAF)**. Still, governments and industry players are investing heavily in SAF development, with ambitious targets set for 2030 and beyond.

Source:) <https://www.thehindu.com/children/fuelling-the-aviation-industry/article69593609.ece>;

<https://www.thehindu.com/news/national/india-to-release-national-policy-on-saf-soon-minister-naidu/article70248674.ece> ;

<https://www.thehindu.com/sci-tech/energy-and-environment/want-a-climate-friendly-flight-its-going-to-take-a-while-and-cost-you-more/article67004879.ece>

Subject:) Environment

Topic:) Energy

Subtopic:)

Q.31)

Ans) a

Exp) Option a is the correct answer.

Statement I is correct: The Government of India has indicated that **from the financial year 2026–27** it intends to **shift the operational fiscal anchor from the fiscal deficit to the debt-to-GDP ratio** as part of its fiscal consolidation strategy.

The government has already set a medium-term fiscal glide path, aiming to reduce the fiscal deficit to about 4.5% of GDP by FY 2025–26, after which the emphasis will gradually move toward managing the overall public debt relative to GDP. Thus, the debt-to-GDP ratio will become the key indicator guiding fiscal prudence and sustainability.

The government's shift is consistent with **global fiscal thinking**, where public debt sustainability is treated as the primary anchor for fiscal prudence rather than annual deficit numbers alone.

Statement II is correct: The **debt-to-GDP ratio** measures the total outstanding government debt as a proportion of the size of the economy. It reflects the **cumulative outcome of past and present borrowing**, unlike **fiscal deficit** which measures **only the government's borrowing requirement in a single year**.

Therefore, it **provides a more comprehensive and long-term indicator of fiscal sustainability**.

Why Statement II explains Statement I:

Because the debt-to-GDP ratio captures the **overall debt burden accumulated over time**, it is viewed internationally as a **better fiscal anchor for assessing fiscal prudence**. This is precisely why the Government of India plans to shift its fiscal consolidation framework towards the debt-to-GDP ratio.

Hence, **both statements are correct and Statement II correctly explains Statement I**.

Source:) <https://www.indiabudget.gov.in/economicsurvey/doc/eschapter/echap02.pdf> Page: 69-70.

<https://www.pib.gov.in/PressReleasePage.aspx?PRID=2221395®=3&lang=1>

<https://www.thehindu.com/opinion/lead/stick-to-fiscal-deficit-as-the-norm-for-fiscal-prudence/article68614653.ece>

<https://www.thehindu.com/business/Economy/indias-fiscal-deficit-for-april-june-at-81-of-202425-target/article68468507.ece>

<https://www.thehindu.com/business/Economy/why-is-fiscal-consolidation-so-important-explained/article67808867.ece>

Subject:) Economy

Topic:) Fiscal Policy

Subtopic:)

Q.32)

Ans) d

Exp) Option d is the correct answer.

Statement I is incorrect: An **increase in the percentage share of gold in India's foreign exchange reserves does not necessarily lead to appreciation of the Indian rupee**.

The value of the rupee in the foreign exchange market is primarily determined by demand and supply of foreign currency, which are influenced by factors such as:

- Balance of payments (exports and imports)
- Capital inflows and outflows (FDI, FPI, external borrowings)
- Inflation and interest rate differentials
- Market expectations and global financial conditions
- Intervention by the Reserve Bank of India

Increasing gold reserves is primarily a **portfolio diversification decision** of the RBI.

While higher gold holdings may improve investor confidence and strengthen external sector resilience, it **does not necessarily cause the Rupee to appreciate**.

Statement II is correct: Gold is widely regarded as a **safe-haven reserve asset** in the global financial system. **Central banks hold gold as part of their reserves because:**

- It is not linked to the economic performance of any single country.
- It helps **diversify foreign exchange reserves**.
- It **reduces excessive dependence on major reserve currencies** such as the US dollar.
- It acts as a **store of value during financial or geopolitical instability**.

Therefore, **diversification of reserves into gold** can **reduce exposure to fluctuations in major currencies** and strengthen the resilience of a country's reserve portfolio.

Hence, **Statement I is not correct but Statement II is correct.**

Source:) <https://www.thehindu.com/business/Economy/how-central-banks-turning-away-from-us-dollar-is-affecting-gold-prices/article70567242.ece>

<https://www.thehindu.com/business/markets/forex-reserves-jump-3293-billion-to-69661-billion/article70464316.ece>

<https://indianexpress.com/article/explained/explained-economics/trump-dedollarisation-gold-surge-10495762/>

Subject:) Economy

Topic:) Monetary Policy

Subtopic:)

Q.33)

Ans) d

Exp) Option d is the correct answer.

Statement I is correct: **Stablecoins** are a type of **blockchain-based digital asset** created with the objective of **maintaining a relatively stable value over time**. Unlike many cryptocurrencies whose prices fluctuate sharply due to market demand and speculation, stablecoins are structured in a way that attempts to keep their value stable, making them more suitable for payments, trading, and digital financial transactions.

Statement II is correct: To maintain stability, the **value of stablecoins is typically pegged to external assets**. These assets may include **fiat currencies** such as the **US Dollar**, **commodities** such as **gold**, or a **basket of assets**. The peg acts as a reference value, helping the stablecoin maintain a relatively constant price in comparison to the referenced asset.

Statement III is correct: In India, **stablecoins** are treated as **part of the broader category of Virtual Digital Assets (VDAs) for taxation purposes**. Under the tax regime introduced in the **Finance Act, 2022**, **income arising from the transfer of VDAs is taxed at a flat rate of 30%**, irrespective of the holding period. Since stablecoins fall within the definition of VDAs, gains from their transfer are also subject to this flat tax rate under the same provisions.

Knowledge Base: **Types of Stablecoins –**

Broadly, **stablecoins can be classified into three types:**

1. Fiat-backed stablecoins – These are backed by reserves of traditional currencies such as the US Dollar or Euro, typically held in banks or regulated financial institutions. Examples include USDT (Tether) and USDC.

2. Crypto-backed stablecoins – These are collateralised by other cryptocurrencies. For example, DAI is backed by crypto assets such as Ethereum.

3. Algorithmic stablecoins – These maintain price stability through automated algorithms that adjust supply and demand rather than relying on reserve backing. They are more experimental and carry higher risk.

Source:) [https://www.thehindu.com/sci-tech/technology/modernisation-of-financial-architecture-how-india-is-adopting-](https://www.thehindu.com/sci-tech/technology/modernisation-of-financial-architecture-how-india-is-adopting-stablecoins/article70134979.ece#:~:text=Stablecoins%20are%20blockchain%2Dbased%20digital,crypto%20assets%20such%20as%20stablecoins)

[stablecoins/article70134979.ece#:~:text=Stablecoins%20are%20blockchain%2Dbased%20digital,crypto%20assets%20such%20as%20stablecoins](https://www.thehindu.com/sci-tech/technology/modernisation-of-financial-architecture-how-india-is-adopting-stablecoins/article70134979.ece#:~:text=Stablecoins%20are%20blockchain%2Dbased%20digital,crypto%20assets%20such%20as%20stablecoins)

<https://www.thehindu.com/business/Economy/union-finance-minister-nirmala-sitharaman-at-kautilya-economic-conclave-2025/article70120196.ece>

<https://www.thehindu.com/business/rbi-deputy-governor-rules-out-stablecoins-in-india/article70388987.ece>

<https://www.thehindu.com/sci-tech/technology/biggest-african-economies-lead-stablecoin-demand-growth-study-shows/article70650224.ece>

Subject:) Economy

Topic:) Digital Currency

Subtopic:)

Q.34)

Ans) b

Exp) Option b is the correct answer.

Option I is correct: Sunyavada

Sunyavada (Doctrine of Emptiness) is a **central philosophical concept of Buddhism**, particularly associated with the **Madhyamaka school founded by Nagarjuna**. It states that all phenomena are empty of inherent or independent existence and arise only through dependent origination (Pratityasamutpada).

Option II is incorrect: Anekantavada

Anekantavada (**Doctrine of Non-absolutism or many-sided reality**) is a **key philosophical principle of Jainism**. It teaches that reality is complex and can be understood from multiple viewpoints, and no single statement can capture the whole truth.

Therefore, **it does not belong to Buddhism**.

Option III is correct: Kshanika-vada

Kshanika-vada (**Doctrine of Momentariness**) is another **important doctrine of Buddhism**. According to this idea, all phenomena exist only for a moment (Kshana) and are constantly arising and perishing. This doctrine supports the Buddhist concept of impermanence (Anitya).

Option IV is incorrect: Nityavada

Nityavada refers to the **doctrine of eternal and unchanging reality, associated with Vedanta within Hinduism**. Vedanta holds that the ultimate reality, Brahman, is eternal (nitya). This idea is contrary to Buddhist doctrines of impermanence and non-self.

Therefore, **it does not belong to Buddhism**.

Thus, only **two of the above doctrines; Sunyavada and Kshanika-vada, primarily belong to Buddhism**.

Source:) " <https://www.egyankosh.ac.in/bitstream/123456789/35230/1/Unit-4.pdf>

<https://egyankosh.ac.in/bitstream/123456789/34935/1/Unit-3.pdf>

<https://egyankosh.ac.in/bitstream/123456789/96131/1/Unit-12.pdf>

<https://egyankosh.ac.in/bitstream/123456789/34657/1/Unit-4.pdf>

<https://egyankosh.ac.in/bitstream/123456789/34676/1/Unit-3.pdf>

Subject:) History

Topic:) Philosophy

Subtopic:)

Q.35)

Ans) c

Exp) Option c is the correct answer.

Information in row I is correct: Poompuhar

Poompuhar (also known as **Kaveripoompattinam**) is **located in Tamil Nadu** at the mouth of the Kaveri River. It was an **important port city of the early Cholas** and finds **mention in Sangam literature and Periplus of the Erythraean Sea**. Marine archaeological explorations have also revealed submerged structural remains near the coast.

Information in row II is correct: Piprahwa

Piprahwa is located in Siddharthnagar district of **Uttar Pradesh**, close to the India-Nepal border. Excavations discovered a **Buddhist stupa** and **relic caskets** containing inscriptions, which are widely believed to contain **relics of the Buddha**. The site is associated with ancient **Kapilavastu**, the capital of the Shakya clan.

Information in row III is correct: Hire Benakal

Hire Benakal in Karnataka is **one of the largest megalithic burial sites in India**. It is particularly famous for its **large number of dolmens** (stone chamber tombs) and other **megalithic burial monuments** such as **menhirs (tall upright stones)**. The site dates back to the **800 BCE to 200 BCE**, making it one of the oldest necropolises in India.

Information in row IV is incorrect: Rangpur

Rangpur is not located in Punjab. It is a **Harappan site in Gujarat** (near the Gulf of Khambhat). Excavations revealed Late Harappan cultural remains and **evidence of rice cultivation**, which is an important feature of the site.

Source:) <https://www.thehindu.com/news/national/tamil-nadu/underwater-archaeological-survey-begins-off-poompuhar-coast-after-more-than-two-decades/article70070562.ece>

<https://www.thehindu.com/news/national/tamil-nadu/yet-another-step-towards-unearthing-the-glorious-past-of-poompuhar-the-capital-of-early-cholas/article69372624.ece>
<https://www.thehindu.com/entertainment/art/piprahwa-buddhist-relics-stupa-lado-sarai-light-and-the-lotus-relics-of-the-awakened-one-naman-ahuja/article70532259.ece>
<https://www.thehindu.com/opinion/editorial/bringing-them-home-on-the-repatriation-of-the-piprahwa-relics/article69920447.ece>
<https://timesofindia.indiatimes.com/travel/destinations/whats-so-special-about-the-megalithic-site-of-hire-benakal-in-karnataka/articleshow/95899921.cms>
<https://egyankosh.ac.in/bitstream/123456789/20133/1/Unit-8.pdf>
<https://egyankosh.ac.in/bitstream/123456789/64780/1/Unit6.pdf>
 Subject:) History
 Topic:) Ancient Sites
 Subtopic:)

Q.36)

Ans) c

Exp) Option c is the correct answer.

Basavanna (also known as Basaveshwara) was a **12th-century social reformer**, poet-saint, and the **founder of the Virashaiva (Lingayat) movement in Karnataka**. He strongly opposed caste hierarchy, ritualism, and Brahmanical orthodoxy, and promoted **devotion to Lord Shiva** in the form of **Ishtalinga**.

- **Basavanna served as a minister (Mahamantri) in the court of Bijjala, a ruler of the Kalachuri Dynasty of Kalyani in north Karnataka (12th century CE).**
- Basavanna held a high administrative position in his court and used his influence to propagate the Lingayat movement.

Source:) <https://egyankosh.ac.in/bitstream/123456789/53714/1/BHIC-132E.pdf> Page: 77.

<https://egyankosh.ac.in/bitstream/123456789/103508/1/Unit-22.pdf> Page: 287.

Subject:) History

Topic:) Philosophy

Subtopic:)

Q.37)

Ans) c

Exp) Option c is the correct answer.

The **Wood's Education Despatch (1854)** was a comprehensive **policy document on education** issued by **Sir Charles Wood**, the **President of the Board of Control**, to the Governor-General of India in 1854.

It is often regarded as the **Magna Carta of English Education in India** because it laid down a systematic framework for the development of modern education in British India.

Statement I is incorrect: The **Wood's Education Despatch did not recommend English at all levels**.

It clearly stated that:

- **Vernacular languages** should be the **medium of instruction** at the **primary level**.
- **English** should be the **medium at the higher levels** of education.

Statement II is correct: Earlier British policy under **Thomas Babington Macaulay**, which assumed that educating a small group of upper-class Indians would naturally "filter down" to the masses.

Wood's Despatch rejected this, emphasizing that the government should take responsibility for the education of the masses.

Statement III is correct: It **recommended the establishment of universities** in the **Presidency towns of Calcutta, Bombay and Madras**, which were set up in **1857** on the model of the London University.

Source:) <https://egyankosh.ac.in/bitstream/123456789/22088/1/Unit-26.pdf> Page: 52.

A History of Modern India, From Plassey to Partition and After by Shekhar Bandyopadhyay Chapter: 3, Early Indian Responses: Reform and Rebellion, Page: 198.

<https://nios.ac.in/media/documents/secichcour/english/ch.18.pdf> Page: 262.

Subject:) History

Topic:) Governance during the British rule

Subtopic:)

Q.38)

Ans) d

Exp) Option d is the correct answer.

Statement I is incorrect: Rekha-prasad, Phamsana and Valabhi are types of Shikhara (superstructure) in the Nagara style of temple architecture. They are not types of Garbhagriha, and they are not associated with the Dravida style.

Statement II is incorrect: A large entrance gateway known as a Gopuram is a characteristic feature of the Dravida architecture, particularly in temples of Tamil Nadu. These towering gateways dominate the temple complex and are enclosed within high boundary walls.

In contrast, the Nagara architecture generally does not feature prominent gopurams; instead, emphasis is placed on the vertical Shikhara above the sanctum.

Source:) <https://ncert.nic.in/textbook/pdf/keks104.pdf> Page: 75.

<https://cuet.iitk.ac.in/sathee-cuet/student-corner/ncert-books/class-11/an-introduction-to-indian-art/chapter-06-temple-architecture-and-sculpture/>

Subject:) Art and Culture

Topic:) Architecture

Subtopic:)

Q.39)

Ans) b

Exp) Option b is the correct answer.

Statement I is incorrect: The World Gold Council is neither an intergovernmental organisation nor was established under the United Nations. It is also not based upon the membership from sovereign nations and international organisations.

The World Gold Council was formed in 1987 by some of the world's most forward-thinking mining companies. It is a membership organisation that champions the role gold plays as a strategic asset, shaping the future of a responsible and accessible gold supply chain. The headquarters of the World Gold Council is located in London, United Kingdom.

- It is a market development organisation/industry association formed by leading gold mining companies.
- Its primary objective is to stimulate and sustain demand for gold, provide market intelligence, and promote responsible gold mining practices.
- It does not regulate international gold trade.

Statement II is correct: Membership of the World Gold Council consists mainly of major global gold-mining companies. It has 29 Members and are some of the world's most forward-thinking gold mining companies. They are headquartered across the world and have mining operations in over 45 countries.

No Indian company is currently a member of the Council. This is largely because India is a major consumer and importer of gold but not a major global gold-mining producer with companies of the scale represented in the Council.

Source:) <https://www.thehindubusinessline.com/multimedia/video/gold-vs-silver-what-the-world-gold-council-says-about-2026/article70428711.ece>

<https://www.gold.org/about-us/our-members>

<https://www.thehindu.com/business/will-indians-still-ride-the-gold-wave-in-2026/article70440982.ece>

<https://indianexpress.com/article/trending/top-10-listing/top-10-countries-with-most-gold-reserves-in-2026-10505490/>

Subject:) International Relations

Topic:) International Organization

Subtopic:)

Q.40)

Ans) a

Exp) Option a is the correct answer.

The **UNESCO World Heritage List** includes cultural, natural and mixed properties of “Outstanding Universal Value” inscribed under the 1972 World Heritage Convention.

India ratified the Convention in 1977 and has since emerged as one of the leading countries in terms of the number of inscribed properties. As of 2025, **India has 44 World Heritage Sites**, comprising **cultural (36), natural (7) and mixed categories (1)**.

Option a is correct: The **latest property from India to be inscribed on the World Heritage List of UNESCO is Maratha Military Landscapes of India**, which was included in 2025 during the 47th session of the World Heritage Committee.

- It became **India's 44th UNESCO World Heritage Site**.
- The property is a serial nomination of **12 historic forts** associated with the Maratha Empire, especially the military system developed under Chhatrapati Shivaji.
- Its **12 components are**; Salher Fort, Shivneri Fort, Lohgad, Khanderi Fort, Raigad, Rajgad, Pratapgad, Suvarnadurg, Panhala Fort, Vijay Durg, and Sindhudurg in Maharashtra, and Gingee Fort in Tamil Nadu.
- This network played a key role in the Marathas' rise as a major political and military force.
- The 'Maratha Military Landscapes' was **developed between the 17th and 19th centuries**.

Option b is incorrect: Sacred Ensembles of the Hoysalas

- **It was inscribed on the World Heritage List of UNESCO in 2023.**
- It includes the Hoysala temples at Belur, Halebidu and Somanathapura in Karnataka.

Option c is incorrect: Moidams – the Mound-Burial System of the Ahom Dynasty

- It was **inscribed on the World Heritage List of UNESCO in 2024.**
- It is located in Assam and associated with the burial traditions of the Ahom rulers.

Option d is incorrect: Santiniketan

- It was **inscribed on the World Heritage List of UNESCO in 2023.**
- It was founded by Rabindranath Tagore as an educational and cultural centre in West Bengal.

Source:) <https://whc.unesco.org/en/statesparties/in>

<https://www.thehindu.com/news/national/indias-maratha-military-landscapes-inscribed-on-unesco-world-heritage-list/article69802028.ece>

Subject:) Art and Culture

Topic:) Miscellaneous

Subtopic:)

Q.41)

Ans) c

Exp) Option c is the correct answer.

Statement I is correct: Since 2011, **Syria has been experiencing a prolonged civil war that began during the Arab Spring** when anti-government protests erupted against the regime of Bashar al-Assad. The Syrian government's violent suppression of these protests escalated the situation into an armed conflict between government forces, opposition groups, and various militant organisations.

Over time, the **conflict expanded into a major geopolitical struggle in West Asia**, with the **involvement of several external actors**; such as Russia and Iran supporting the Syrian government, while countries like Turkey and the United States supported different opposition or Kurdish groups. Thus, **Syria became a focal point of regional and global strategic rivalry**.

Statement II is incorrect: In 2011, the President of Syria was not removed through a military coup. Instead, Bashar al-Assad continued to remain in power while the country descended into civil war following protests and armed rebellion. **Assad was removed from power when rebels overthrew his regime in December, 2024**, forcing him to flee the country. The conflict was triggered by popular protests and subsequent repression, not by a military coup or the formation of an interim military administration.

Therefore, **Statement I is correct but Statement II is not correct.**

Source:) <https://www.thehindu.com/news/international/as-violence-and-instability-spread-in-syria-isis-is-reorganising-and-regaining-strength-rohilat-afrin/article70582797.ece>

<https://www.thehindu.com/podcast/global-faultlines-podcast-understanding-the-syrian-crisis-what-led-the-country-into-civil-war/article70379977.ece>

<https://www.thehindu.com/news/international/whats-happening-in-syria-explained/article68946549.ece>

<https://www.thehindu.com/opinion/editorial/sectarian-wounds-on-the-violence-in-syria/article69846860.ece>

Subject:) International Relations

Topic:) Places in News

Subtopic:)

Q.42)

Ans) a

Exp) Option a is the correct answer.

Statement I is correct: A reduction in the Cash Reserve Ratio (CRR) by the Reserve Bank of India may induce commercial banks to lower their lending interest rates. When liquidity in the banking system increases, banks have greater capacity to extend credit. With improved liquidity conditions and reduced funding constraints, banks may lower lending rates to expand credit demand and remain competitive.

Statement II is correct: CRR refers to the percentage of a bank's Net Demand and Time Liabilities (NDTL) that must be maintained as reserves with the RBI in cash form. When the CRR is reduced, banks are required to keep a smaller proportion of their deposits with the RBI. This directly increases the funds available with banks for lending and investment purposes.

How Statement II Explains Statement I

A decrease in CRR → increases lendable resources with banks → enhances liquidity in the banking system → increases supply of credit → may lead to reduction in lending interest rates.

Thus, **Statement II provides the mechanism that explains why a reduction in CRR can induce banks to lower their lending rates.**

Therefore, **both statements are correct and Statement II explains Statement I.**

Source:) <https://www.thehindu.com/business/Economy/rbi-cuts-crr-by-a-steep-1-to-unlock-25-lakh-crore-to-bank-funds-by-december/article69664016.ece>

<https://www.pib.gov.in/PressNoteDetails.aspx?NoteId=154573&ModuleId=3>

https://www.thehindu.com/opinion/columns/C_R_L__Narasimhan/CRR-still-serves-a-purpose/article12640270.ece

Subject:) Economy

Topic:) Monetary Policy

Subtopic:)

Q.43)

Ans) c

Exp) Option c is the correct answer.

Statement I is correct: Under Article 87 of the Constitution of India, the President addresses both Houses of Parliament assembled together at the beginning of the first session each year and after every general election to the Lok Sabha.

After this address, a Motion of Thanks is moved separately in each House; the Lok Sabha and the Rajya Sabha, to express gratitude for the President's Address. The motion is then debated in both Houses.

Statement II is incorrect: Amendments to the Motion of Thanks can be moved in both Houses of Parliament, not only in the Lok Sabha.

Members of the Lok Sabha as well as the Rajya Sabha may propose amendments highlighting criticisms of government policies or omissions in the President's Address.

Statement III is correct: As per the Lok Sabha Secretariat, **the discussion on the Motion of Thanks is concluded by the reply of the Prime Minister or any other Minister.** The mover or the seconder of the motion does not have any right of reply at the end unlike other motions. Immediately thereafter, the amendments are disposed of and the Motion of Thanks is put to vote.

Similarly in Rajya Sabha, at the end of the discussion, the Prime Minister replies to the debate. Thereafter, the amendments are disposed of, and then the Motion of Thanks is put to vote in the House. If any of the amendments is accepted then the Motion of Thanks is adopted in the amended form.

Source:) Mastering Indian Polity by Forum IAS, Chapter:16 - Legislative Branch, Page: 389.

<https://www.thehindu.com/news/national/lok-sabha-passes-resolution-motion-of-thanks-on-president-address-pm-address/article70594740.ece>

<https://www.thehindu.com/news/national/congress-flags-procedural-irregularities-in-motion-of-thanks-debate/article70597083.ece>

https://cms.rajyasabha.nic.in/UploadedFiles/Procedure/PracticeAndProcedure/English/4/president_address.pdf

https://sansad.in/uploads/president_e_157317094a.pdf?updated_at=2022-09-13T09:09:58.126Z

Subject:) Polity

Topic:) Legislature

Subtopic:)

Q.44)

Ans) c

Exp) Option c is the correct answer.

The **procedure for amendment of the Constitution** of India is **provided under Article 368** of the Constitution of India.

- The Constitution is neither completely rigid nor completely flexible; it follows a partly rigid and partly flexible amendment procedure.
- **Constitutional amendments** can be **initiated only through a Constitution Amendment Bill in Parliament**, which must be passed with a special majority, and in certain cases also requires ratification by at least half of the State Legislatures before receiving the President's assent.

Statement I is incorrect: A **Constitution Amendment Bill** under Article 368 **does not require prior recommendation of the President for its introduction in Parliament.**

It can be introduced in either House of Parliament (Lok Sabha or Rajya Sabha) by a minister or a private member without any prior approval from the President.

Statement II is incorrect: In case of **disagreement between the two Houses, no provision exists for a joint sitting for a Constitution Amendment Bill.**

Although joint sittings can be convened under **Article 108** of the Constitution of India for ordinary legislation, this provision does not apply to Constitution Amendment Bills. Each House must independently pass the bill with the required special majority.

Statement III is correct: The **President is bound to give assent to a Constitutional Amendment Bill** once it is **duly passed by Parliament** following the procedure under **Article 368.**

Unlike ordinary bills, the President:

- Cannot withhold assent, and
- Cannot return the Amendment Bill for reconsideration.

This position was clarified after the **24th Constitutional Amendment Act, 1971.**

Source:) Mastering Indian Polity by Forum IAS, Chapter:4 – Constitutional Amendment & Basic Structure Doctrine, Page: 22.

<https://www.mea.gov.in/images/pdf1/part20.pdf>

Subject:) Polity

Topic:) Constitutional Amendment

Subtopic:)

Q.45)

Ans) c

Exp) Option c is the correct answer.

President's Rule (State Emergency) refers to the situation when the constitutional machinery in a State fails and the Union government assumes control of the State administration. It is provided under **Article 356** of the Constitution of India.

It can be **proclaimed by the President** of India, usually on the basis of a report from the State Governor or otherwise, when the **government of a State cannot be carried on in accordance with the provisions of the Constitution**.

A **proclamation of President's Rule** issued under Article 356 of the Constitution of India **must be approved by both Houses of the Parliament** of India **within two months from the date of its issuance**.

If the Lok Sabha is dissolved at the time, the proclamation survives until 30 days from the first sitting of the reconstituted Lok Sabha, provided the Rajya Sabha has approved it in the meantime. If the proclamation is not approved within this period, it automatically ceases to operate. Once approved, it remains in force for six months and may be extended with further parliamentary approval.

Statement I is incorrect: While **Punjab** was the **first state to see President's Rule (1951)**, it is **not the state where it has been imposed the maximum number of times**. That distinction belongs to **Manipur (11 times)**, followed closely by **Uttar Pradesh (10 times)**.

However, these are not the states (including UTs) that have spent the most time under central control. That distinction is held by Jammu & Kashmir, followed by Punjab and Puducherry.

Statement II is correct: Once President's Rule is in effect, the President dismisses the state council of ministers. Additionally, the **President may declare** that the **powers of the State Legislature shall be exercisable by or under the authority of the Parliament of India**.

Statement III is incorrect: The **Constitution permits the President's Rule to continue for a maximum period of three years**.

The Constitution permits extension beyond one year only if:

1. A National Emergency under Article 352 is in operation in the whole or any part of the State; and
 2. The Election Commission certifies that President's rule is necessary due to difficulties in conducting State elections.
- However, even with these conditions, the maximum permissible duration is three years.

Source:) Mastering Indian Polity by Forum IAS, Chapter: 14 – Emergency, Page: 248.

<https://www.thehindu.com/news/national/how-does-a-presidents-rule-function-explained/article69227389.ece>

<https://indianexpress.com/article/explained/explained-law/presidents-rule-manipur-what-is-the-provision-history-9830266/>

<https://www.thehindu.com/news/national/how-is-presidents-rule-imposed-explained/article69650172.ece>

<https://www.thehindu.com/news/national/manipur-six-years-under-president-rule-infoographics/article69411784.ece>

Subject:) Polity

Topic:) Emergency

Subtopic:)

Q.46)

Ans) b

Exp) Option b is the correct answer.

Statement I is correct: The IUCN periodically evaluates the conservation status of natural World Heritage sites through its **World Heritage Outlook assessment**. Under this framework, sites are categorized into **four broad status groups**: Good, Good with Some Concerns, Significant Concern, and Critical.

- In its **recent World Heritage Outlook 4 (2025) report**, the **Western Ghats** (a UNESCO natural World Heritage site inscribed in 2012) has been **placed in the category of “Significant Concern.”**
- This classification indicates that the site’s natural values are under considerable pressure and that existing conservation measures may not be sufficient to safeguard them in the long term.
- This is due to **high threats from factors** like **climate change**, hydropower projects, tourism pressures, invasive species, forest loss, urbanization, and developmental activities, despite ongoing conservation efforts.

Statement II is correct: The **Western Ghats** were **inscribed as a natural World Heritage Site in 2012** because of their **Outstanding Universal Value (OUV)**. The Western Ghats have Outstanding Universal Value which is manifested in the region’s unique and fascinating influence on large-scale biophysical and ecological processes over the entire Indian peninsula. The region is globally recognized for:

- **Exceptional biodiversity**
- Very **high degree of endemism** (many species found nowhere else in the world)
- Important **ecological and climatic functions**, including their role in influencing the Indian monsoon system and maintaining hydrological stability across peninsular India.

However, Statement II does not explain Statement I.

Statement I refers to the current conservation status evaluation of the site by the IUCN (i.e., the site is under “Significant Concern”).

Statement II explains why the Western Ghats were originally recognized as a World Heritage Site, due to their ecological importance and biodiversity.

Therefore, while both statements are correct, Statement II does not explain the classification mentioned in the Statement I.

Hence, **Both Statement I and Statement II are correct but Statement II does not explain Statement I.**

Source: <https://www.thehindu.com/sci-tech/energy-and-environment/why-has-iucn-red-flagged-the-western-ghats-explained/article70178487.ece>

<https://whc.unesco.org/en/list/1342/>

Subject:) Environment

Topic:) Miscellaneous

Subtopic:)

Q.47)

Ans) a

Exp) Option a is the correct answer.

Statement I is correct: In recent years, **tensions and periodic armed confrontations along the border between Thailand and Cambodia** have **resulted in a humanitarian situation in the border regions**.

- Military exchanges in contested areas have affected nearby civilian settlements, forcing many residents to leave their homes and move to safer areas or temporary shelters.
- Such clashes have **disrupted livelihoods**, damaged infrastructure, and created **large-scale internal displacement and humanitarian distress** among local populations.

Statement II is correct: The **two countries share a long and historically disputed border**. A **major point of contention** has been the **territory surrounding the Preah Vihear Temple**, the **11th century Khmer Hindu temple** and **UNESCO World Heritage Site** located on the frontier. Although the temple itself was awarded to Cambodia by the **International Court of Justice** in **1962**, disagreement has persisted over adjacent land and boundary demarcation, largely due to ambiguities in colonial-era maps.

Link between the statements:

The **unresolved territorial dispute around Preah Vihear and nearby border areas** periodically leads to military standoffs and armed clashes between Thailand and Cambodia. These confrontations occur in populated border zones, which **directly results in civilian displacement and humanitarian problems**.

Hence, **Both Statement I and Statement II are correct and Statement II explains Statement I**.

Source:) " [https://www.thehindu.com/news/international/the-view-from-india-newsletter-why-thailand-and-cambodia-are-](https://www.thehindu.com/news/international/the-view-from-india-newsletter-why-thailand-and-cambodia-are-fighting/article70401923.ece#:~:text=Bangkok%20accused%20Cambodia%20of%20mining,full%2Dscale%20cross%20border%20fighting)

[fighting/article70401923.ece#:~:text=Bangkok%20accused%20Cambodia%20of%20mining,full%2Dscale%20cross%20border%20fighting](https://www.thehindu.com/news/international/the-view-from-india-newsletter-why-thailand-and-cambodia-are-fighting/article70401923.ece#:~:text=Bangkok%20accused%20Cambodia%20of%20mining,full%2Dscale%20cross%20border%20fighting)

<https://www.thehindu.com/news/international/why-have-thailand-cambodia-border-clashes-flared-up-again-explained/article70408717.ece> .

<https://www.thehindu.com/news/international/over-500000-displaced-in-cambodia-by-thai-border-clashes-interior-ministry/article70421990.ece>

<https://www.who.int/publications/m/item/public-health-situation-analysis---cambodia---thailand-border-conflict#:~:text=Overview,military%20buildup%20on%20both%20sides>

Subject:) International Relations

Topic:) Places in News

Subtopic:)

Q.48)

Ans) c

Exp) Option c is the correct answer.

The **ADEETIE Scheme** (Assistance in Deploying Energy Efficient Technologies in Industries & Establishments) is a **Central Sector scheme of the Ministry of Power**.

- It aims to **promote the adoption of energy-efficient technologies among Micro, Small and Medium Enterprises (MSMEs)**, particularly in energy-intensive industrial clusters.
- The scheme will be implemented over a period of **three years**, starting from **Financial Year (FY) 2025-26** and ending in **FY 2027-28**.

Statement I is incorrect: The ADEETIE scheme is an **initiative of the Ministry of Power, not the Ministry of New and Renewable Energy (MNRE)**. It is **implemented by the Bureau of Energy Efficiency (BEE)**, a statutory body under the Ministry of Power.

Statement II is correct: The **scheme provides financial support through interest subvention on loans** to Micro, Small and Medium Enterprises for Energy Efficiency technology adoption. Specifically, it offers a **5% interest subvention for Micro and Small Enterprises** and a **3% interest subvention for Medium Enterprises**, ensuring accessibility and affordability for MSMEs seeking financial aid for energy efficiency (EE) projects.

Statement III is correct: To ensure focused implementation, the **scheme is targeted at MSMEs registered under the Udyam portal and operating within identified industrial clusters of certain identified sectors**.

MSMEs registered on the Udyam portal and operating in the **60 identified energy-intensive clusters** of the following **14 sectors are eligible**:

- Brass, Bricks, Ceramics, Chemicals, Fisheries, **Food Processing**, Forging, Foundry, Glass & Refractory, Leather, Paper, Pharmaceutical, Steel Re-Rolling, **Textiles**.

Source:) <https://www.thehindu.com/news/national/telangana/centre-offers-1000-crore-to-msmes-for-energy-efficiency-initiatives/article69883449.ece>

<https://www.pib.gov.in/PressReleasePage.aspx?PRID=2144822®=3&lang=2>

<https://adeetie.beeindia.gov.in/operation-guideline-of-scheme>

<https://beeindia.gov.in/adeetie.php>

Subject:) Economy

Topic:) Government Schemes

Subtopic:)

Q.49)

Ans) c

Exp) Option c is the correct answer.

The CROPIC (Collection of Real Time Observations and Photographs of Crops) initiative is a **technology-driven digital program** launched by the **Ministry of Agriculture and Farmers Welfare**, Government of India, under the **Pradhan Mantri Fasal Bima Yojana** (PMFBY), the country's flagship crop insurance scheme.

It is a **digital application for collection of crop field photographs**, to ascertain the health of crops and identify damage due to perils.

Statement I is incorrect: The CROPIC initiative is a technology-driven program **launched under the Pradhan Mantri Fasal Bima Yojana** (PMFBY), which is India's flagship crop insurance scheme aimed at providing financial support to farmers against crop losses.

It is not associated with the Pradhan Mantri Krishi Sinchayi Yojana (PMKSY), which focuses on irrigation and water management in agriculture.

CROPIC (Collection of Real Time Observations and Photographs of Crops), one of the technology initiatives under Pradhan Mantri Fasal Bima Yojana (PMFBY), **is a digital application for collection of crop field photographs, to ascertain the health of crops and identify damage due to perils.** CROPIC app aims at collection of geotagged crowdsourced photographs from farmers and field coordinators through smartphones. The purpose of timeseries geotagged photographs is crop validation against insured crops, and crop damage assessment in case of localized calamities.

Statement II is incorrect: The initiative is not funded through the **Agriculture Infrastructure Fund** (AIF). The AIF primarily finances post-harvest infrastructure projects such as warehouses, cold chains, and agri-logistics.

In contrast, **CROPIC is financed through the Fund for Innovation and Technology** (FIAT) created under PMFBY. The FIAT has a **total outlay of Rs 825 crore** for different technology innovations under crop insurance schemes.

Source:) <https://www.pib.gov.in/PressReleasePage.aspx?PRID=2148514®=3&lang=2>

<https://www.pib.gov.in/PressReleasePage.aspx?PRID=2148516>

https://sansad.in/getFile/annex/268/AU1454_BFHyZc.pdf?source=pqars

<https://indianexpress.com/article/explained/new-scheme-to-study-crops-using-ai-what-is-cropic-10060949/>

Subject:) Economy

Topic:) Government Schemes

Subtopic:)

Q.50)

Ans) c

Exp) Option c is the correct answer.

PM-SETU (Pradhan Mantri Skilling and Employability Transformation through Upgraded ITIs) is a flagship **centrally-sponsored initiative** launched by the Government of India in **October 2025** with an outlay of **₹60,000 crore**. It aims to enhance the overall quality and relevance of vocational training in the country.

Statement I is incorrect: PM-SETU is a **Centrally Sponsored Scheme** with funding shared among the **Central Government (₹30,000 crore)**, **State Governments (₹20,000 crore)**, and **industry partners (₹10,000 crore)**. It is not fully funded by the Union Government as a Central Sector Scheme.

Statement II is correct: The **PM-SETU Scheme aims to upgrade 1,000 Government Industrial Training Institutes (ITIs) across India into modern, industry-aligned training institutions.** The objective is to modernize infrastructure, improve training quality, and align vocational education with the evolving needs of industry so that trainees gain skills relevant to the labour market.

A unique feature of PM-SETU is its **industry-led governance**. Each upgraded ITI under the scheme **will be managed by a Special Purpose Vehicle (SPV)**, which is a partnership where **industry leaders hold 51% ownership** and the **government holds 49%**. This co-investment model allows companies to lead the transformation while receiving up to 83% in government co-funding.

Statement III is correct: PM-SETU will follow a hub-and-spoke model, with 200 hub ITIs linked to 800 spoke ITIs. Each hub will be equipped with advanced infrastructure, innovation and incubation centres, production units, training of trainer facilities, and placement services, while the spokes will extend access and outreach.

Knowledge Base: The **key objectives of the PM-SETU scheme are:**

1. To improve the quality of training delivery in Industrial Training Institutes (ITIs) and NSTIs;
2. To modernize infrastructure and equipment as per industry standards;
3. To introduce industry-aligned long-term and short-term courses, especially in new and emerging sectors;
4. To strengthen industry linkage for demand-driven skilling and better employment outcomes; and
5. To enhance the capacity of National Skill Training Institutes (NSTIs) for training of trainers. The scheme includes **strengthening five selected National Skill Training Institutes (NSTIs)** in Bhubaneswar, Chennai, Hyderabad, Kanpur, and Ludhiana **as global Centres of Excellence**, incorporating global partnerships for advanced training and trainer development.

Source:) <https://www.pib.gov.in/PressReleaseDetailm.aspx?PRID=2208158®=3&lang=1>

<https://www.pib.gov.in/PressReleasePage.aspx?PRID=222121®=3&lang=1>

<https://www.india.gov.in/spotlight/details/pradhan-mantri-skilling-and-employability-transformation-through-upgraded-itis-pm-setu>

<https://dgt.gov.in/sites/default/files/2025-10/PM-SETU-Guidelines-Component-I-Upgradation-of-ITIs.pdf>

Subject:) Economy

Topic:) Government Schemes

Subtopic:)

Q.51)

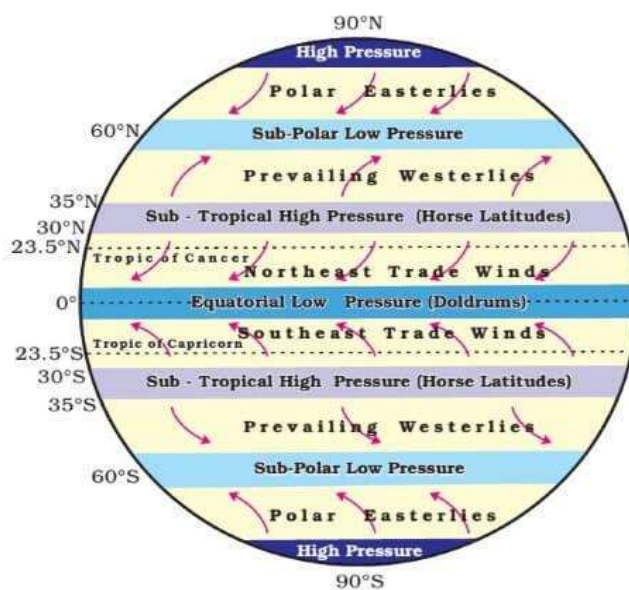
Ans) c

Exp) Option c is the correct answer.

The Earth's permanent wind system is controlled by global pressure belts and the rotation of the Earth. Air always moves from high pressure to low pressure regions. However, due to the Coriolis force, the direction of these winds gets deflected. This gives rise to three major planetary winds – Trade Winds, Westerlies and Polar Easterlies.

Statement I is correct. The term **anti-trade** literally means “**opposite to the trade winds.**” The **westerlies** are called anti-trade winds **because they blow in a direction opposite to the trade winds.** While trade winds move from east to west in the tropics, the westerlies move from west to east in the mid-latitudes.

Statement II is incorrect. While trade winds blow from the sub-tropical high-pressure belts towards the equator, the westerlies originate from the **sub-tropical high pressure belts around 30° latitude (not polar high pressure belts).** They move towards the sub-polar low pressure belts around 60° latitude. Due to the **Coriolis force, they are deflected west to east.** This makes their direction opposite to that of the trade winds.



Major Pressure Belts and Wind System

The movement of winds from sub-tropical highs to sub-polar lows, combined with Coriolis deflection, results in a west-to-east flow. That west-to-east movement is the key reason for the name anti-trade winds as it is opposite to trade winds flow.

Source: <https://egyankosh.ac.in/bitstream/123456789/29518/1/Unit-1.pdf> - Unit 1 - The Hydrologic Cycle and Hydrometeorology - Atmospheric Circulation - Major Planetary Winds - Page 22

<https://www.egyankosh.ac.in/bitstream/123456789/112324/1/Unit-9.pdf>

<https://ncert.nic.in/textbook/pdf/gess204.pdf>

Subject:) Geography

Topic:) Planetary Winds

Subtopic:)

Q.52)

Ans) b

Exp) Option b is the correct answer.

The troposphere is the lowest layer of the atmosphere and almost all weather phenomena occur in this layer. One of its most important characteristics is the **normal lapse rate**, that is, the decrease in temperature with height. The temperature in the troposphere generally decreases with increasing height. The rate of decrease of temperature with height is termed as the normal lapse rate. It is per .

Statement I is correct. The temperature decreases with altitude in the troposphere because the atmosphere is **heated mainly from below**. The Earth's surface absorbs incoming solar radiation. It then re-radiates energy in the form of **long-wave terrestrial radiation**. This outgoing radiation heats the air in contact with the surface. As we move upward, we go farther from this primary source of heating. Therefore, the air becomes cooler with height. This is the fundamental reason behind the normal lapse rate in the troposphere.

Statement II is correct. The Earth does receive energy from the Sun in the form of **short-wave solar radiation**. This is the ultimate source of energy for the entire Earth-atmosphere system. Without this incoming radiation, neither the surface nor the atmosphere would be heated.

However, most of this short-wave radiation passes through the atmosphere and directly heats the Earth's surface rather than the atmosphere itself. The atmosphere is mainly heated due to long wave radiations from the surface of the Earth.

Statement II does not explain Statement I. While short-wave solar radiation is the original energy source, it does not directly explain the decrease of temperature with altitude in the troposphere. The lapse rate exists because the atmosphere is heated from the Earth's surface through terrestrial radiation, conduction and convection. The mechanism of vertical temperature decrease depends on **heating from below**, not simply on the arrival of solar energy.

Source: NCERT - 11th Fundamental of Physical Geography - Chapter 8 - Composition and Structure of Atmosphere - Structure of the Atmosphere - Page 73 and Chapter 9 - Solar Radiation, Heat Balance and Temperature - Solar Radiation - Page 75 and Heating And Cooling of Atmosphere - Page 76

Subject:) Geography

Topic:) Climatology

Subtopic:)

Q.53)

Ans) a

Exp) Option a is the correct answer.

The Earth's interior cannot be directly observed because of extreme depth, temperature and pressure. Therefore, scientists depend on both **direct evidence** and **indirect scientific methods**. Direct sources involve actual material that can be physically examined. Indirect sources rely on interpretation of physical properties such as wave behaviour or gravity variation.

Option I is correct. Mining provides actual rock samples from beneath the surface. Although the depth reached is limited to a few kilometres, the material collected is physically examined in laboratories. Hence it is a **direct source** of information.

Option II is incorrect. Meteors do not originate from the Earth's interior. Their composition is studied because it resembles that of the Earth. However, they only provide comparative evidence. They are **Indirect sources** of Information.

Option III is correct. Volcanic eruptions eject molten material from within the Earth. Once solidified, this material is available for laboratory examination. Although the exact depth of origin may not always be known, the material itself comes from inside the Earth. Hence, this is a **direct source** of information.

Option IV is incorrect. Gravity measurements do not provide physical samples. They indicate variations in mass distribution within the Earth. Scientists interpret these anomalies to understand internal structure. This method depends on inference, not material collection. Therefore, it is an **indirect source** of Information.

Option V is incorrect. Seismic waves travel through the Earth and change speed depending on the density and state of materials. Their study has helped identify layers like crust, mantle and core.

However, this method relies on interpretation of wave behaviour. It does not provide direct material samples. Hence, it is an **indirect source** of Information.

Source: <https://ncert.nic.in/textbook/pdf/kegy203.pdf> - NCERT - 11th Fundamental of Physical Geography – Chapter 3 – Interior of the Earth – Sources of Information About The Interior – Page 18

Subject:) Geography

Topic:) Geomorphology

Subtopic:)

Q.54)

Ans) d

Exp) Option d is the correct answer.

Tides are among the most predictable movements of ocean water. They are governed by celestial mechanics but have deep ecological and administrative consequences. From shaping marine biodiversity in the intertidal belt to influencing disaster policy in vulnerable coastal states, tides are not merely a physical process.

Statement I is correct. Ocean tides occur due to the **gravitational pull of the Moon and the Sun on the Earth's surface**. The Moon plays the dominant role because it is much closer to the Earth, even though the Sun is more massive. This gravitational attraction causes periodic rise and fall of sea levels.

Statement II is correct. The intertidal zone lies between the high tide mark and the low tide mark. In many regions, especially those with semi-diurnal tides, this area experiences two high tides and two low tides daily. Organisms living in this zone must survive alternating exposure to air and submergence in water. They face rapid changes in salinity, temperature and oxygen availability. Over time, many species have evolved structural and physiological adaptations to survive these harsh and shifting conditions.

Statement III is correct. Kerala has recently declared coastal high-tide flooding or sea incursion as a State-specific disaster. This allows compensation to affected residents under norms similar to the State Disaster Response Fund. It is the **first such declaration in India** addressing recurrent tidal flooding as a disaster category.

According to a government order, since tidal rise is a routine natural phenomenon, a normal tidal rise cannot be declared a disaster under State Disaster Response Fund (SDRF) norms, making it difficult to provide financial assistance to those affected under the norms of the SDRF. However, Section 2(d) of the Disaster Management Act provides that if flooding caused by tidal rise adversely affects a community and results in loss of life, livelihood, and living conditions, such a natural phenomenon can be considered a disaster. Making use of this provision, Kerala now notifies 'coastal high-tide flooding/sea incursion' as a State-specific disaster and lakhs of affected people will be benefited from this.

Source: <https://ncert.nic.in/textbook/pdf/gess205.pdf> Chapter 5 - Water - Tides - Page 36

<https://ncert.nic.in/textbook/pdf/kegy213.pdf> - Movements of Ocean Water – Tides – Page 109

<https://www.thehindu.com/news/national/kerala/why-kerala-declared-tidal-flooding-a-state-specific-disaster-explained/article70641866.ece>

<https://www.thehindu.com/children/between-the-tides/article70253486.ece>

Subject:) Geography

Topic:) Tides

Subtopic:)

Q.55)

Ans) c

Exp) Option c is the correct answer.

Soybean production is highly concentrated in a few countries. According to recent trade data and global agricultural reports, **Brazil and the United States of America together dominate global soybean production as well as exports.**

Brazil has emerged as the **largest soybean producer and exporter in the world.** Its production has expanded significantly due to large-scale cultivation in regions such as Mato Grosso and the Cerrado.

The **United States is the second-largest producer**, with major cultivation in Midwest states like Iowa and Illinois. The U.S. has traditionally been a leading exporter, though it now faces competition from Brazil.

Argentina is a major producer and exporter but ranks below Brazil and the United States in total production volume. China, although the largest importer of soybeans, is **not among the top two producers.** India's production is significant domestically but much smaller compared to the global leaders.

Source: <https://www.thehindubusinessline.com/opinion/from-soyabeans-to-strategic-alliances-a-usindia-opportunity/article70175030.ece> ; <https://www.thehindubusinessline.com/economy/agri-business/brazils-soy-trap-more-land-falling-yields-and-rising-climate-risk/article70519072.ece> ; <https://www.thehindu.com/business/by-the-numbers-breaking-down-the-global-soybean-trade/article69237682.ece> ;

<https://worldpopulationreview.com/country-rankings/soybean-production-by-country>

Subject:) Geography

Topic:) Crops

Subtopic:)

Q.56)

Ans) a

Exp) Option a is the correct answer.

The Indus river system forms one of the largest drainage systems in the world. In India, its major Himalayan tributaries originate in different latitudinal zones of Jammu & Kashmir and Himachal Pradesh before flowing into Pakistan.

Jhelum rises from a spring at **Verinag** at the foot of the **Pir Panjal range** in the Kashmir Valley. It flows through **Srinagar** and **Wular Lake** before entering Pakistan. Among the listed rivers, it is situated the **northernmost**, within India.

Chenab is formed by the union of the **Chandra and Bhaga** streams at Tandi in Himachal Pradesh. It flows through the Jammu region before entering Pakistan. It lies **south of the Jhelum**, but north of the Ravi and Sutlej.

Sutlej originates near **Raksas Tal** close to Mansarovar in Tibet. After flowing parallel to the Indus for some distance, it enters India through **Shipki La** and then moves into the Punjab plains. Among the four listed rivers, it is the **southernmost** in India.

Therefore, the correct North to South arrangement is:

- **Jhelum**
- **Chenab**
- **Sutlej**

Knowledge Base:



Source: NCERT - 11th India Physical Environment - Drainage System - The Indus System - Page 24

Subject: Geography

Topic: Drainage System

Subtopic:

Q.57)

Ans) c

Exp) Option c is the correct answer.

India has two distinct coastal plains i.e. the Western Coastal Plains along the Arabian Sea and the Eastern Coastal Plains along the Bay of Bengal. Though both are important for maritime trade, their geomorphological character differs significantly.

Statement I is correct. The western coast provides better natural conditions for the development of ports and harbours than the eastern coast. The coastline along the Arabian Sea is indented and possesses natural inlets. Such features allow ships to anchor safely without requiring heavy artificial construction. This is why major natural harbours like Mumbai and Kochi developed along this coast.

Because of its emergent nature and more continuous coastline, the eastern coast of India has less number of ports and harbours.

Statement II is correct. The western coast of India is largely a **submergent coast**. Submergent coasts are formed due to subsidence or relative rise in sea level. They are characterised by drowned river valleys and indented shorelines. In contrast, the eastern coast is mainly an **emergent coast**, where depositional features dominate and the coastline is comparatively straighter.

Statement II explains Statement I. The key reason for better natural harbour development on the western coast lies in its **submergent character**. Submergence leads to indented coastlines and natural inlets, which are ideal for port development. Although delta formation on the eastern coast is a correct geographical feature, it does not directly explain the superior harbour conditions on the western coast. The decisive factor is the submergent nature of the western shoreline.

Statement III is incorrect. The eastern coastal rivers such as the Mahanadi, Godavari, Krishna and Kaveri form extensive **deltas** before entering the Bay of Bengal. On the western side, rivers are shorter and steeper due to the proximity of the Western Ghats. They usually form **estuaries** instead of large deltas.

Source: NCERT - 11th India Physical Environment - Structure and Physiography - The Coastal Plains - Page 14

Subject: Geography

Topic: Indian Physiography

Subtopic:

Q.58)

Ans) d

Exp) Option d is the correct answer.

The **Persian Gulf** is a strategically vital water body located between the Arabian Peninsula and Iran. It is central to global energy security, as nearly one-third of the world's seaborne crude oil trade passes through it, particularly via the Strait of Hormuz.



Recent tensions between Iran and Israel, and Iran-US, have once again highlighted the strategic importance of the **Persian Gulf**. The Gulf is virtually an enclosed sea with the only opening being in the east through the Strait of Hormuz. **Eight states border the Persian Gulf: Bahrain, Iran, Iraq, Kuwait, Oman, Qatar, Saudi Arabia and the United Arab Emirates.** The Strait of Hormuz remains a geopolitical chokepoint, and any instability in the region directly impacts global crude oil prices and shipping insurance premiums.

Option I is correct. **Bahrain** is an island nation located entirely within the Persian Gulf. It has no land boundary with Iran but lies off the eastern coast of Saudi Arabia. It directly borders the Persian Gulf.

Option II is correct. **Kuwait** is situated at the north-western tip of the Persian Gulf. It shares maritime boundaries with both Iraq and Saudi Arabia and has direct access to the Gulf.

Option III is correct. Iraq has a small, 58 km (36 miles) coastline at the northern end of the Persian Gulf. This limited coastline is located on the Al-Faw Peninsula between Kuwait and Iran.

Option IV is correct. **Qatar** is a peninsula projecting into the Persian Gulf. It is surrounded on three sides by the Gulf and shares a land boundary only with Saudi Arabia.

Option V is correct. **Saudi Arabia** has a long eastern coastline along the Persian Gulf, in addition to its western coastline along the Red Sea.

Source: <https://www.indiatoday.in/information/story/what-are-gcc-countries-the-gulf-states-caught-between-iran-and-the-us-israel-war-2876401-2026-03-02> ; <https://www.thehindu.com/news/national/kerala/israel-iran-conflict-keralite-expatriates-in-gulf-face-heightened-anxiety-and-fear-amid-escalating-tensions-in-west-asia/article70694285.ece> ; <https://www.indiatoday.in/world/story/oil-tanker-with-15-indians-on-board-hit-off-oman-coast-as-israel-iran-exchange-fresh-attacks-4-injured-2876167-2026-03-01>

Subject:) Geography

Topic:) Mapping

Subtopic:)

Q.59)

Ans) a

Exp) Option a is the correct answer.

Tiger Reserves in India are designated under **Project Tiger**, which was launched in 1973 to ensure the conservation of the Bengal Tiger and its habitat. These reserves are governed under the **Wildlife (Protection) Act, 1972**, and function on a **core-buffer strategy**.

- The **core area** is kept inviolate for wildlife conservation.
- The **buffer area** permits regulated human activities to reduce pressure on the core.

Importantly, many Tiger Reserves are structured around **major river systems**, which sustain forest ecology, provide perennial water sources, and support biodiversity.

Row I is incorrect. **Panna Tiger Reserve** is located in **Madhya Pradesh**, and the **Ken River** flows through it, forming spectacular gorges and sustaining its dry deciduous forest ecosystem.

Row II is correct. **Satkosia Tiger Reserve** is situated in **Odisha** and is centred around the gorge formed by the **Mahanadi River**, which is ecologically and geographically integral to the reserve.

Row III is incorrect. **Achanakmar Tiger Reserve** lies in **Chhattisgarh**. The **Maniyari River**, a tributary of the **Mahanadi basin**, flows through the reserve area and contributes significantly to its forest hydrology and biodiversity.

Row IV is incorrect. **Mudumalai Tiger Reserve** is located in **Tamil Nadu**, forming part of the Nilgiri Biosphere landscape. The **Moyar River** flows through the reserve and creates the well-known Moyar Gorge, playing a vital ecological role in the region.

Source: <https://forest.mponline.gov.in/eBrochure/eBrochureDetailsV2.aspx?parkid=3> ;
<https://ntca.gov.in/assets/uploads/briefnote/satkoshia.pdf> ; <https://achanakmartigerreserve.com/> ;
<https://ntca.gov.in/assets/uploads/briefnote/mudumalai.pdf>
 Subject:) Geography
 Topic:) Tiger Reserves
 Subtopic:)

Q.60)

Ans) a

Exp) Option a is the correct answer.

Mountain ranges are among the most prominent physical features of the Earth's surface. They are formed through tectonic processes such as **continental collision, volcanic activity, and crustal uplift**. Major mountain systems like the **Himalayas, Andes and Alps along with Rockies** are vital for various highest peaks world over.

Row I is incorrect. **Mount Everest** is located in the **Himalayas** and is situated on the Nepal-China border (border of Nepal and Tibet Autonomous Region of China and not in India). It is the world's highest peak.

Row II is correct. **Aconcagua** is the highest peak of the **Andes Mountains** and is located in **Argentina**. It is also the highest mountain outside Asia.

Row III is incorrect. **Mont Blanc** is the highest peak of the **Alps**, and it is primarily associated with **France** not Canada. Mont Blanc lies on the France-Italy border, with surrounding areas in Switzerland.

Row IV is incorrect. **Mount Elbert** is the highest peak of the **Rocky Mountains**, and it is located in the **United States (Colorado)**, not Switzerland.

Source: <https://indianexpress.com/article/technology/science/why-does-mount-everest-keep-growing-9596786/> ;
<https://indianexpress.com/article/india/tripura-engineer-mt-aconcagua-bengaluru-andes-10503387/> ;
<https://indianexpress.com/article/news-archive/print/air-india-plane-crash-46-years-later-indian-diplomatic-bag-discovered-on-mt-blanc/> ; <https://www.worldatlas.com/mountains/rocky-mountains.html>
 Subject:) Geography
 Topic:) World Physiography
 Subtopic:)

Q.61)

Ans) c

Exp) Option c is the correct answer.

The organisms **Tenkana jayamangali**, **Piratula acuminata**, and **Mimetus spinatus** are spider species that were recently reported in biodiversity research in India.

- **Tenkana jayamangali** is a species of jumping spider belonging to the family Salticidae. It was discovered in **Karnataka** and represents part of a newly identified genus of spiders from southern India.
- **Piratula acuminata** is a wolf spider of the family Lycosidae, reported from the **Sundarbans region**. Its discovery also marked the first record of the genus *Piratula* in India.
- **Mimetus spinatus** belongs to the family Mimetidae, commonly known as pirate spiders, which are known for preying on other spiders. This species was discovered in the **Western Ghats**.

Source:) <https://www.thehindu.com/sci-tech/energy-and-environment/new-genus-of-jumping-spiders-tenkana-discovered-in-south-india/article68783276.ece>

<https://www.thehindu.com/sci-tech/energy-and-environment/zsi-scientists-discover-a-wolf-spider-and-four-wasp-species-from-west-bengal/article69861712.ece>

<https://indianexpress.com/article/cities/kolkata/zoological-survey-of-india-discovery-of-2-spider-species-in-western-ghats-9544792/>

Subject:) Environment

Topic:) Biodiversity

Subtopic:)

Q.62)

Ans) a

Exp) Option a is the correct answer.

Statement I is correct: Studies and expert reports have highlighted that **black plastic** products used in food containers, takeaway packaging, and **kitchen utensils can leach harmful chemicals into food**. The risk increases when such containers are exposed to heat (for example, microwave use), oils, or prolonged contact with food. Such migration of chemicals from plastic to food can **pose potential health risks**.

Statement II is correct: **Black plastic** is frequently manufactured using **recycled plastic waste**, including material sourced from **discarded electronic equipment (e-waste)**. Electronic plastics, especially in older electronics products, often **contain additives** such as **brominated flame retardants** and may also contain **heavy metals**. When such materials are recycled into food-contact products without adequate purification, these hazardous substances can remain in the final product.

Link between Statement I and Statement II:

Statement II explains Statement I. The presence of brominated flame retardants and heavy metals in recycled black plastic is the underlying reason why harmful chemicals may leach into food. Thus, the chemical composition of recycled e-waste plastic explains the health risk mentioned in Statement I.

Therefore, **Both Statement I and Statement II are correct and Statement II explains Statement I.**

Source:) <https://indianexpress.com/article/explained/explained-sci-tech/black-plastic-9854309/>

<https://frontline.thehindu.com/news/kitchen-food-safety-black-plastic-utensils-toxic-leaching-chemicals-recycling/article69018970.ece>

Subject:) Environment

Topic:) Pollution

Subtopic:)

Q.63)

Ans) b

Exp) Option b is the correct answer.

Pair I is correct: AIM4NatuRe Initiative

The **Food and Agriculture Organization (FAO)** launched the **AIM4NatuRe** (Accelerating Innovative Monitoring for Nature Restoration) **Initiative** to strengthen countries' capacity to monitor and report ecosystem restoration.

The initiative supports implementation of targets under the **Kunming-Montreal Global Biodiversity Framework** by improving data, tools and methodologies for tracking restoration of forests, wetlands, grasslands and other ecosystems.

Pair II is incorrect: Global Energy Alliance for People and Planet (GEAPP)

The GEAPP is a **multi-stakeholder global partnership** created to **accelerate the clean-energy transition** and expand energy access in developing countries.

It was launched with major support from philanthropic institutions such as the **Rockefeller Foundation**, **IKEA Foundation**, and **Bezos Earth Fund**, along with development finance partners.

It is **not led by the World Bank**, though multilateral institutions may collaborate with it.

Pair III is correct: Net-Zero Nature-Positive Accelerator Integrated Program

The **UN Environment Programme (UNEP)** and the **Global Environment Facility (GEF)**, together with the **Development Bank of Latin America (CAF)** and the **Asian Development Bank (ADB)**, have officially **launched the Net-Zero Nature-Positive Accelerator Integrated Programme**.

The initiative seeks to align climate and biodiversity goals by driving systemic economic and financial transformation, while mobilizing governments, businesses, and communities to take urgent, integrated action.

Source:) <https://www.fao.org/newsroom/detail/fao-UK-aim4nature-to-ecosystem-restoration/en>

<https://www.thehindu.com/business/india-needs-to-mobilise-500-billion-a-year-to-meet-net-zero-ambition-global-energy-alliance/article70644201.ece>

<https://energyalliance.org/who-we-are/>

<https://www.unep.org/news-and-stories/press-release/unep-launches-100-million-programme-accelerate-climate-and-nature>

Subject:) Environment

Topic:) Climate Change Initiatives

Subtopic:)

Q.64)

Ans) c

Exp) Option c is the correct answer.

Statement I is incorrect: The **Third UN Ocean Conference (UNOC3)** was **co-hosted by France and Costa Rica**, and it was **held in Nice, France** (9–13 June 2025).

The earlier **Second UN Ocean Conference (2022)** was **co-hosted by Portugal and Kenya** in Lisbon, which adopted the **Lisbon Declaration**.

All UN Ocean Conferences, including the 2025 edition, are specifically designed to **support the implementation of Sustainable Development Goal 14 (Life Below Water)**. The 2025 theme was **"Accelerating action and mobilizing all actors to conserve and sustainably use the ocean."**

Statement II is incorrect: While the **High Seas Treaty**, formally known as the **BBNJ (Biodiversity Beyond National Jurisdiction) Agreement**, was a major talking point at the 2025 Conference, **it was not formally adopted for the first time during this Conference**.

The **treaty reached its required 60th ratification (by Morocco) in September 2025**, and **officially entered into force on January 17, 2026**.

During the Nice conference in June 2025, many countries signed or ratified it, but the threshold for "entry into force" had not yet been met.

Statement III is correct: The **primary outcome of the 2025 Conference** was the **adoption of the "Nice Ocean Action Plan."** This plan includes a high-level **political declaration** titled **"Our Ocean, Our Future: United for Urgent Action"** and **voluntary commitments** aimed at strengthening marine conservation, reducing ocean pollution, expanding marine protected areas, and enhancing ocean finance.

Source:) <https://www.thehindu.com/sci-tech/energy-and-environment/what-was-decided-at-the-un-oceans-conference-explained/article69710425.ece>

<https://www.thehindu.com/news/national/kerala/shipping-disasters-off-kerala-coast-highlighted-at-un-ocean-meet-in-nice/article69683780.ece>

<https://sdgs.un.org/conferences/ocean2025>

<https://press.un.org/en/2025/sea2231.doc.htm>

<https://www.un.org/bbnjagreement/en>

Subject:) Environment

Topic:) Climate Change Initiatives

Subtopic:)

Q.65)

Ans) d

Exp) Option d is the correct answer.

The **Gangetic River Dolphin** (*Platanista gangetica*), popularly known as the **Susu**, is a **freshwater cetacean native to the river systems** of the **Indian subcontinent**. It inhabits the **Ganga-Brahmaputra-Meghna and Karnaphuli-Sangu river systems** across India, Nepal, and Bangladesh.

It is classified as **Endangered** by the **IUCN** and is listed under **Schedule I** of the **Wildlife (Protection) Act, 1972**, providing it the highest level of legal protection in India. In 2009, it was declared the **National Aquatic Animal of India** to strengthen conservation efforts.

Statement I is correct: The **Gangetic River Dolphin** is **blind**. Its eyes lack a well-developed crystalline eye lens, so it cannot form clear images.

II. It **navigates and hunts using echolocation**, i.e., emitting high-frequency ultrasonic sounds that bounce off objects and prey.

III. This adaptation helps it survive in the muddy, turbid waters of large rivers.

Statement II is correct: The species is mostly **solitary in behaviour**.

- **Individuals** are usually **seen alone** or in **small groups**.
- Typically, a **mother and calf move together**, which is the most common grouping observed.

Statement III is incorrect: In Gangetic River Dolphins, **females are generally larger than males**, which is an example of **sexual dimorphism** in the species.

Statement IV is correct: The **Vikramshila Gangetic Dolphin Sanctuary** (VGDS), located in the Bhagalpur district of **Bihar**, remains the **only notified sanctuary specifically dedicated** to the **conservation of this freshwater dolphin** in India. It covers roughly 50 km of the Ganga River.

Source:) <https://www.worldwildlife.org/species/dolphin-and-porpoise/ganges-river-dolphin/>

<https://www.thehindu.com/sci-tech/energy-and-environment/first-estimate-of-gangetic-dolphins-finds-6327-across-four-states/article69286367.ece>

<https://indianexpress.com/article/lifestyle/pets-animals/indias-pink-secret-why-the-ganges-is-the-only-place-to-see-the-amazons-long-lost-counterpart-10483895/>

<https://timesofindia.indiatimes.com/city/patna/vikramshila-gangetic-dolphin-sanctuary-a-last-bastion-for-endangered-dolphins-in-bihar/articleshow/121852966.cms>

<https://indianexpress.com/article/explained/ganges-river-dolphin-project-dolphin-conservation-9734143/>

Subject:) Environment

Topic:) Biodiversity

Subtopic:)

Q.66)

Ans) b

Exp) Option b is the correct answer.

Option a is incorrect: The Global Carbon Harvest Coalition/Alliance was officially launched during the COP30 summit of UNFCCC in Belém, Brazil (November 2025), as part of the "Plan to Accelerate Solutions" (PaS) for carbon removal. It was **not launched during COP29 of UNFCCC**, which was held in Baku, Azerbaijan in 2024.

Option b is correct: The Global Carbon Harvest Coalition is a high-profile South-South and North-South collaborative initiative **led by Brazil, India, and Kenya**.

Option c is incorrect: The coalition's primary goal is to **promote soil-based carbon removal and climate resilience through agriculture**. It focuses on scientific validation and standardizing MRV (Monitoring, Reporting, and Verification) for techniques like **biochar application** and enhanced **rock weathering**. It is a **voluntary initiative, not a "legally binding global carbon tax."**

Option d is incorrect: The coalition is **not funded exclusively by the Green Climate Fund (GCF)**. It is a collaborative platform involving governments, research institutions and other partners, and its financing is expected to come from multiple sources such as **public funding, climate finance and private investment, rather than a single dedicated fund**.

Source:) <https://www.carbonbusinesscouncil.org/news/cop30>

https://climateaction.unfccc.int/assets/documents/26_.pdf Page: 3.

Subject:) Environment

Topic:) Climate Change Initiatives

Subtopic:)

Q.67)

Ans) c

Exp) Option c is the correct answer.

As per the findings of the **India State of Forest Report 2023** published by the **Forest Survey of India**:

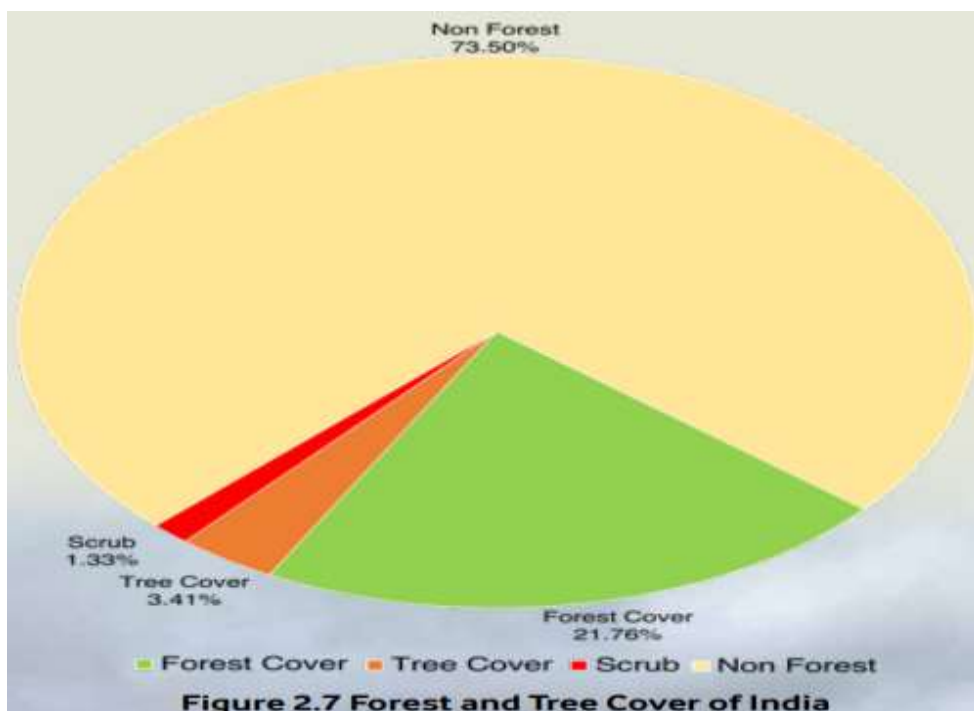
- The total forest and tree cover of India is **8,27,356.95 sq km**, which is **25.17%** of the geographical area of the country.
 - **Forest Cover: 7,15,342.61 sq km (21.76%)**
 - **Tree Cover: 1,12,014.34 sq km (3.41%)**
- The total forest and tree cover of the country has **increased by 1,445.81 sq km** compared to the 2021 assessment.
- The **maximum increase in forest and tree cover** has been observed in the States of **Chhattisgarh** (683.62 sq km) followed by **Uttar Pradesh** (559.19 sq km), **Odisha** (558.57 sq km) and Rajasthan (394.46 sq km).
- The **maximum decrease in forest and tree cover** has been noticed in the state of **Madhya Pradesh** (612.41 sq km) followed by Karnataka (459.36 sq km), Ladakh (159.26 sq km) and Nagaland (125.22 sq km).

Thus, **three states (Odisha, Uttar Pradesh, and Chhattisgarh)** among the given options recorded a **net increase in forest and tree cover**.

Hence, the **correct answer is: c) Only three**

Knowledge Base: **India State of Forest Report 2023:**

Table: Forest and Tree Cover of India		
		in km ²
Class	Area	Percentage of GA
Forest Cover	7,15,342.61	21.76
Tree Cover	1,12,014.34	3.41
Total Forest and Tree Cover	8,27,356.95	25.17
Scrub	43,622.64	1.33
Non Forest	24,16,489.29	73.50
Geographical Area of the country	32,87,468.88	100.00



Source: <https://www.pib.gov.in/PressReleasePage.aspx?PRID=2086742®=3&lang=2>

https://fsi.nic.in/uploads/isfr2023/isfr_book_eng-vol-1_2023.pdf

Subject: Environment

Topic: Biodiversity

Subtopic:

Q.68)

Ans) d

Exp) Option d is the correct answer.

India has submitted the nomination dossier titled “*Jingkieng Jri / Lyu Chrai Cultural Landscape*” to UNESCO for inscription on the World Heritage List. The nomination represents the unique living root bridges located in the Khasi and Jaintia Hills of Meghalaya.

- These bridges are **bio-engineered structures** created by training the aerial roots of the **rubber fig tree** (*Ficus elastica*) across streams and rivers.
- They are developed and maintained by **indigenous Khasi and Jaintia communities** using traditional ecological knowledge passed down through generations.
- The site has been proposed as a cultural landscape, recognizing the **harmonious interaction between humans and nature**, and the sustainable engineering practices embedded in local traditions.

Source: <https://www.thehindu.com/news/national/india-submits-nomination-of-meghalayas-living-root-bridges-for-unesco-world-heritage-site/article70564187.ece>

<https://www.newsonair.gov.in/india-submits-meghalayas-living-root-bridges-to-unesco-for-world-heritage-status/>

Subject: Environment

Topic: Biodiversity

Subtopic:

Q.69)

Ans) b

Exp) Option b is the correct answer.

In an ecosystem, organisms do not live in isolation but interact continuously with one another. These interactions play a crucial role in maintaining ecological balance and shaping community structure.

Based on the effects on the interacting species, **ecological interactions are broadly classified into:**

- **Mutualism (+ / +):** Both species benefit (e.g., mycorrhizae in plant roots).
- **Commensalism (+ / 0):** One species benefits while the other remains unaffected (e.g., remora fish and shark).
- **Amensalism (- / 0):** One species is harmed while the other is unaffected (e.g., antibiotic secretion by *Penicillium*).
- **Parasitism (+ / -):** One benefits and the other is harmed.
- **Predation (+ / -):** One organism kills and consumes another.
- **Competition (- / -):** Both species are negatively affected due to resource rivalry.

Commensalism:

Commensalism is a type of ecological interaction in which **one species benefits while the other remains unaffected (+/0 interaction)**. The benefiting organism may gain food, shelter, support, or transportation, without causing harm or providing benefit to the host species.

Option I is correct: Remora fish attaching to sharks for transportation and leftover food:

The remora benefits by obtaining free transport and feeding on leftover scraps from the shark's meals. The shark is neither harmed nor benefited by the remora's presence.

Hence, **this is a classic example of commensalism (+/0 interaction).**

Option II is incorrect: Mycorrhizae (fungi) associated with the roots of higher plants:

In mycorrhizal association, the fungus enhances water and mineral (especially phosphorus) absorption for the plant, while the plant supplies carbohydrates to the fungus.

Since **both organisms derive benefit, this represents mutualism (+/+ interaction) and not commensalism.**

Option III is incorrect: *Penicillium* secreting antibiotics that inhibit bacterial growth:

The fungus *Penicillium* secretes antibiotics that inhibit or kill nearby bacteria. The bacteria are harmed, while the fungus does not gain direct benefit from this inhibition.

This interaction is categorized as amensalism (-/0 interaction).

Option IV is correct: Orchids growing as an epiphyte on a mango branch

Epiphytic orchids use the mango tree only for physical/structural support to access better sunlight. They do not derive nutrients from the tree and do not harm it. The orchid benefits, while the host tree remains unaffected.

Thus, **it is commensalism (+/0 interaction).**

Source:) <https://ncert.nic.in/textbook/pdf/lebo111.pdf> Page: 197 & 201.

<https://www.egyankosh.ac.in/bitstream/123456789/12758/1/Unit-11.pdf> Page: 89.

Subject:) Environment

Topic:) Ecology and Ecosystem

Subtopic:)

Q.70)

Ans) c

Exp) Option c is the correct answer.

Option I is correct: Gharial (*Gavialis gangeticus*)

The Gharial is classified as Critically Endangered on the IUCN Red List of Threatened Species. Its population has declined sharply due to river habitat degradation, **sand mining**, dam construction, and accidental **entanglement in fishing nets**.

The Gharial is a **freshwater crocodilian** inhabiting the river systems of northern India, particularly the Ganga, Chambal, and their tributaries. It is primarily fish-eating and is distinguished by its **long, narrow snout**.

Option II is correct: Great Indian Bustard (*Ardeotis nigriceps*)

The Great Indian Bustard is classified as Critically Endangered. The species has suffered catastrophic decline due to habitat loss of grasslands, agricultural expansion, and, most significantly, **mortality from collisions with overhead power transmission lines**.

It is **one of the heaviest flying birds** and is **endemic to the Indian subcontinent**. It inhabits **arid and semi-arid grasslands** of India, mainly in Rajasthan and parts of Gujarat.

Option III is incorrect: Asiatic Lion (*Panthera leo persica*)

The Asiatic Lion is categorized as Endangered, not Critically Endangered.

The Asiatic Lion is a subspecies of lion found mainly in and around the **Gir landscape of Gujarat**, particularly in **Gir National Park**.

Due to sustained conservation efforts, including habitat protection and community involvement, its population has shown a steady increase.

Option IV is correct: Bengal Florican (*Houbaropsis bengalensis*)

The Bengal Florican is listed as Critically Endangered. Rapid loss of grassland habitats due to agricultural expansion and human activities has pushed the species to the Critically Endangered category.

The Bengal Florican is a **rare bustard species inhabiting tall grasslands of the Indian subcontinent** and Southeast Asia. In India, it occurs mainly in the grasslands in the Gangetic and Brahmaputra floodplains.

Source: <https://frontline.thehindu.com/environment/successful-conservation-breeding-india-biodiversity/article70188907.ece>

<https://timesofindia.indiatimes.com/city/meerut/depleted-river-dependent-iucns-first-gharial-green-status-assessment-calls-for-river-protection-to-save-species/articleshow/124421785.cms>

<https://indianexpress.com/article/explained/out-of-the-woods-in-gujarat-why-asiatic-lions-future-cant-be-secured-by-numbers-alone-10022963/>

<https://www.iucnredlist.org/species/8966/149227430>

<https://www.iucnredlist.org/species/22691932/134188105>

<https://www.iucnredlist.org/species/247279613/259031465>

<https://www.iucnredlist.org/species/22692015/130184896>

Subject: Environment

Topic: Biodiversity

Subtopic:)

Q.71)

Ans) b

Exp) Option b is the correct answer.

Dedicated Freight Corridors (DFCs) are high-capacity railway corridors designed exclusively for freight movement, aimed at decongesting existing railway networks and boosting logistics efficiency. The two major corridors under implementation are the **Eastern DFC (EDFC)** and the **Western DFC (WDFC)**, executed by the Dedicated Freight Corridor Corporation of India Ltd. (DFCCIL).

Options II and III are correct and options I and IV are incorrect. The **Eastern Dedicated Freight Corridor (EDFC)** runs from **Sahnewal (Punjab)** to **Dankuni (West Bengal)**. It passes through the following states **Punjab, Haryana, Uttar Pradesh, Bihar, Jharkhand and West Bengal**. Thus, it passes through a total of **six states**.

The **Western Dedicated Freight Corridor (WDFC)** runs from **Dadri (Uttar Pradesh)** to **Jawaharlal Nehru Port (Maharashtra)**. It traverses the following states

Uttar Pradesh, Haryana, Rajasthan, Gujarat and Maharashtra. Thus, it passes through a total of **five states**.

Hence, Both the Eastern Dedicated Freight Corridor (EDFC) and the Western Dedicated Freight Corridor (WDFC) pass through two common states i.e. Haryana and Uttar Pradesh.

Source: <https://www.thehindu.com/news/national/dedicated-freight-corridor-registers-47-increase-in-train-trips-in-2024-2025/article69995906.ece> ;

<https://www.pppinindia.gov.in/assets/doc/20181/22330/106th%20PPPAC%20Meeting.pdf>

<https://economictimes.indiatimes.com/industry/transportation/shipping/-transport/jawaharlal-nehru-port-connectivity-of-dedicated-freight-corridor-by-march-31-2026-official/articleshow/126466949.cms?from=mdr>

https://mtp.indianrailways.gov.in/view_detail.jsp?lang=0&id=0,4,268&dcd=804&did=160913740949836846154413CF25895A69042B0FE5B26

Subject:) Geography

Topic:) Human and Economic Geography

Subtopic:)

Q.72)

Ans) c

Exp) Option c is the correct answer.

The Inter-Tropical Convergence Zone (ITCZ) is one of the most crucial components of the Earth's atmospheric circulation system. It forms part of the broader framework of **pressure belts and planetary wind systems** that regulate global climate patterns. Its seasonal movement plays a decisive role in redistributing heat and moisture across the tropics and has profound implications for monsoon behaviour, rainfall variability, and tropical weather systems.

Statement I is correct and statement II is incorrect: Near the equator, the Sun's rays are almost vertical throughout the year. This causes **intense heating of the land and oceans**. Warm air becomes lighter and rises (not descends). When air rises, it creates a **low-pressure zone** at the surface. The **north-east trade winds (from Northern Hemisphere)** and **south-east trade winds (from Southern Hemisphere)** move toward this low-pressure area and converge. This zone of convergence and rising air is called the **ITCZ**.

Hence, the Inter Tropical Convergence Zone (ITCZ) is a low pressure zone located at the equator where trade winds converge, and so, it is a zone where air tends to ascend.

Statement III is correct. The ITCZ does not stay fixed at the equator. It moves because the **area of maximum heating shifts with the Sun**. In **June-July (northern hemisphere summer)**, the Sun is overhead near the Tropic of Cancer. The landmass of Asia heats up strongly. The ITCZ shifts **northward**. Whereas, in **December-January (southern hemisphere summer)**, the Sun is overhead near the Tropic of Capricorn. The ITCZ shifts **southward**. Thus, the ITCZ follows the **apparent movement of the Sun** shifting northward during the Northern Hemisphere summer and southward during the Southern Hemisphere summer.

Source: <https://egyankosh.ac.in/bitstream/123456789/104276/3/Unit-9.pdf> - Unit 9 - Monsoon - Origin of Monsoon With Special Reference to India - Page 154;

NCERT - 11th Fundamental of Physical Geography - Atmospheric Circulation and Weather Systems - General circulation of the atmosphere - Page 88

<https://ncert.nic.in/textbook/pdf/kegy104.pdf> Page 30

Subject:) Geography

Topic:) Climatology

Subtopic:)

Q.73)

Ans) b

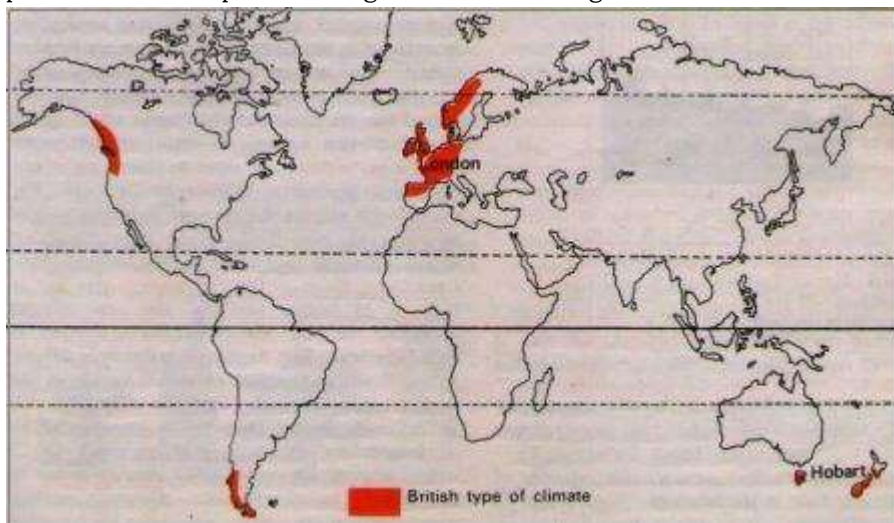
Exp) Option b is the correct answer.

The climatic features and vegetation clearly indicate that it is a **British Type or The Cool Temperate Western Margin Climate**.

Statement I. The British Type climate is found on the **western margins of continents in the mid-latitudes**. It occurs in regions such as Western Europe, western Canada and parts of southern Chile. These areas lie between 40° and 60° latitudes and face the oceans directly, allowing maritime influence throughout the year.

Statement II. The region experiences a **moist climate with precipitation throughout the year**, although rainfall is relatively higher in winter. Summers are mild and do not become completely dry. This distinguishes it from the Mediterranean climate, where summers are distinctly dry and winters are the only rainy season. In the British type climate, cyclonic rainfall associated with temperate depressions occurs frequently.

Statement III. The British Type climate region remains under the **permanent influence of the Westerlies**. These prevailing winds bring moist air masses from the oceans, resulting in regular rainfall. The natural vegetation consists of **temperate deciduous hardwood forests**, including Oak, Elm, Ash, Beech, Maple and Chestnut. These trees often grow in pure stands with sparse undergrowth and have high commercial value for timber.



Knowledge Base:

- The **Mediterranean climate** is found on the western margins of continents, but it is characterised by **hot, dry summers and mild, wet winters**. The presence of dry summers is its defining feature.
- The **humid subtropical climate** occurs mainly on the **eastern margins of continents**, such as southeastern USA and eastern China. It experiences hot summers and rainfall throughout the year, often with summer highs due to convection and cyclones.
- The **steppe climate** is a **semi-arid climate** found in the interior of continents. It receives low and unreliable rainfall and supports grasslands rather than temperate deciduous forests.

Source: <https://egyankosh.ac.in/bitstream/123456789/103169/3/Unit-15.pdf> - Unit 15 - Climatic Regions of the World - Climatic Classification - Cool Temperate Zone - Page 258;

Certificate Physical and Human Geography by G C Leong - Chapter on The Cool Western Margin (British Type) Climate Subject:) Geography

Topic:) Climatology

Subtopic:)

Q.74)

Ans) c

Exp) Option c is the correct answer.

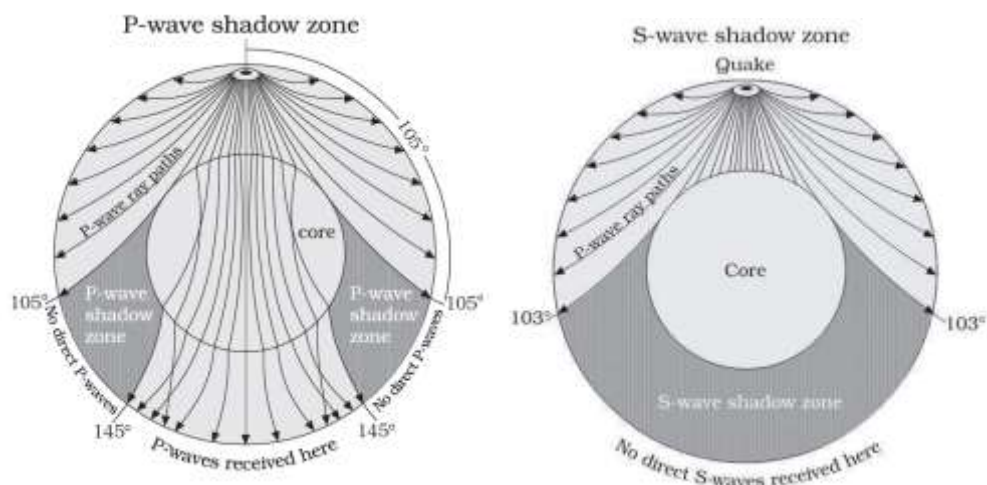
An **earthquake** is the sudden shaking of the Earth's surface caused by the rapid release of energy within the lithosphere, usually due to the movement of tectonic plates along faults. The point inside the Earth where energy is released is called the **focus (hypocentre)**, and the point directly above it on the surface is known as the **epicentre**.

Earthquakes are measured in two important ways:

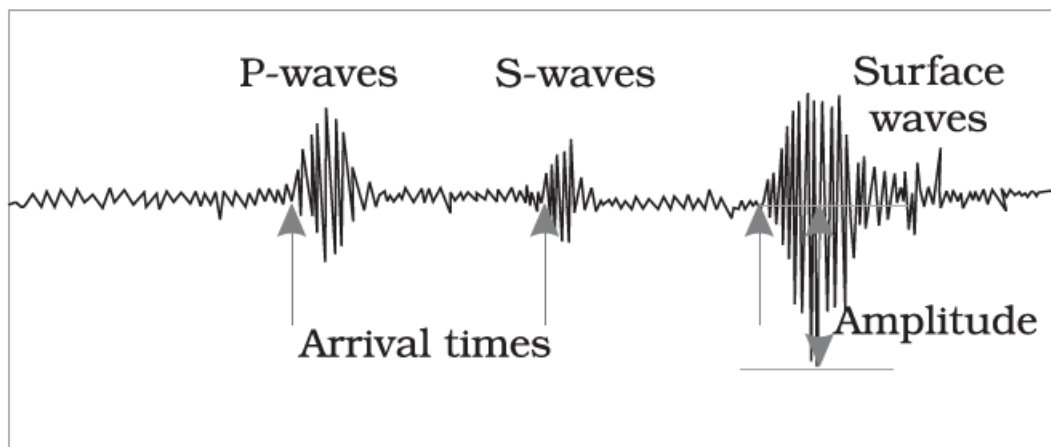
- **Magnitude**, which indicates the amount of energy released. It is measured using the **Richter Scale**.
- **Intensity**, which refers to the observed effects and damage caused by the earthquake at a particular location. It is measured using the **Mercalli Scale**.

Statement I is correct. P-waves (Primary waves) are compressional waves that can travel through **solids, liquids and gases**. This property allows them to pass through both the Earth's mantle and outer core. Whereas, S-waves (Secondary waves), however, can travel **only through solid materials**. They cannot move through liquids. This is why S-waves do not pass through the Earth's liquid outer core.

Seismographs show that both P and S waves are recorded up to **105° from the epicentre**. Beyond 105°, S-waves are not recorded at all, creating a large **S-wave shadow zone**. P-waves, due to refraction in the liquid outer core, form a distinct **shadow zone between 105° and 145°**. Beyond 145°, P-waves reappear.



Statement II is correct. Surface waves are the **last among seismic waves to be recorded on the seismograph**. They arrive **after P and S waves** but are the most destructive because they move along the Earth's surface and cause maximum ground displacement. Thus compared to body waves i.e. P and S waves, surface waves are more destructive.



Source: NCERT - 11th Fundamental of Physical Geography - Interior of The Earth - Earthquake Waves - Page 22
 Subject:) Geography
 Topic:) Earthquake
 Subtopic:)

Q.75)

Ans) b

Exp) Option b is the correct answer.

The **Coriolis force** is an apparent force caused by the **rotation of the Earth**. It deflects moving objects to the **right in the Northern Hemisphere** and to the **left in the Southern Hemisphere**. It acts only on bodies that are in motion relative to the Earth's surface and plays a crucial role in atmospheric and oceanic circulation.

Option I is incorrect. The length of the day is determined by the **rotation of the Earth on its axis**. While the Coriolis force arises because of Earth's rotation, it does not influence how long the Earth takes to complete one rotation. Therefore, the length of day is **not affected** by the Coriolis force. Coriolis force mainly affects/influences the moving objects.

Option II is incorrect. Seasons occur due to the **tilt of the Earth's axis (23½°)** and its revolution around the Sun. The Coriolis force has no role in determining seasonal changes. Therefore, seasons of Earth are **not affected** by the Coriolis force.

Option III is correct. Trade winds are large-scale planetary winds that blow from subtropical high-pressure belts toward the equatorial low-pressure belt. Due to the Coriolis force, these winds are deflected:

- To the right in the Northern Hemisphere (forming north-east trades)
- To the left in the Southern Hemisphere (forming south-east trades)

Thus, trade winds are **significantly influenced** by the Coriolis force.

Option IV is correct. Major ocean currents are set in motion primarily by winds, and their direction is strongly modified by the Coriolis force. This deflection leads to the formation of **clockwise gyres in the Northern Hemisphere** and **anticlockwise gyres in the Southern Hemisphere**. Therefore, ocean currents are **significantly affected** by the Coriolis force.

Option V is correct. The Coriolis force is essential for the **rotation of tropical cyclones**. It causes cyclones to rotate:

- Counterclockwise in the Northern Hemisphere
- Clockwise in the Southern Hemisphere

In fact, cyclones do not form very close to the equator because the Coriolis force is weak there. Hence, tropical cyclones are **significantly affected**.

Source: NCERT - 11th Fundamental of Physical Geography - Atmospheric Circulation and Weather Systems - Forces Affecting the Velocity and Direction of Wind - Page 86

<https://digital.nios.ac.in/content/316en/ch11.pdf> - Page 211

Subject:) Geography

Topic:) Climatology

Subtopic:)

Q.76)

Ans) d

Exp) Option d is the correct answer.

Article 6.4 of the Paris Agreement establishes a **UNFCCC mechanism** for generating and trading high-quality carbon credits. India, through the **National Designated Authority for Implementation of the Paris Agreement (NDAIPA)**, has notified a list of eligible GHG mitigation activities for participation under bilateral and cooperative approaches.

Option a is correct. **Green Hydrogen** is produced using renewable energy and does not emit carbon during production. It helps in decarbonising hard-to-abate sectors such as steel and fertilisers. It is an approved mitigation activity under Article 6.4.

Option b is correct. **Offshore wind energy** projects generate clean electricity and significantly reduce fossil fuel dependence. They are capital-intensive but align with India's low-carbon transition goals and are eligible for carbon credit trading under Article 6.4.

Option c is correct. India has included Compressed bio-gas power in its notified list of GHG mitigation activities under the Article 6.4 mechanism.

Option d is incorrect. **Agriculture, Forestry and Other Land Use (AFOLU) sectors** include activities such as afforestation, agroforestry, sustainable land management and restoration of degraded land. Despite their role in carbon sequestration and rural livelihoods, they are currently not eligible under India's notified list for carbon credit trading under Article 6.4.

Source:<https://unfccc.int/process-and-meetings/the-paris-agreement/article6>

https://moef.gov.in/uploads/pdf/article_6.4.pdf

<https://indianexpress.com/article/opinion/columns/indias-climate-policy-must-not-leave-farmers-behind-10338538/>

Subject:) Environment

Topic:) Pollution

Subtopic:)

Q.77)

Ans) d

Exp) Option d is the correct answer.

Coral bleaching has become one of the clearest biological warning signs of climate change. The ongoing global mass bleaching event, impacts on the Great Barrier Reef, and severe stress in Indian reef systems such as Lakshadweep highlight the growing vulnerability of marine ecosystems to rising ocean temperatures.

Statement I is correct. Coral bleaching occurs when corals, under environmental stress, particularly heat stress, **expel zooxanthellae**, the symbiotic algae living in their tissues. These algae provide food through photosynthesis and give corals their colour. **Once expelled, the white calcium carbonate skeleton becomes visible, giving the coral a bleached appearance. Bleached corals are not immediately dead but are physiologically weakened.**

Statement II is correct. The primary trigger of coral bleaching is **elevated sea surface temperature**, especially when temperatures remain about 1°C above the seasonal maximum for several weeks. **Marine heatwaves**, intensified by global warming and El Niño events, are major drivers. Accumulated heat stress is measured using the **Degree Heating Week (DHW)** index, and higher DHW values are strongly associated with mass bleaching.

Statement III is correct. The global mass bleaching event that began in 2023 has been recorded as the **largest on record**, affecting nearly 84% of the world's coral reef area. This includes the **Great Barrier Reef**, the world's largest coral reef system. The scale and geographic spread of this event underscore the increasing frequency and intensity of climate-induced marine heat stress worldwide.

Knowledge Base:



Source: <https://indianexpress.com/article/explained/coral-bleaching-9967400/> ; <https://www.thehindu.com/sci-tech/energy-and-environment/great-barrier-reef-records-largest-annual-coral-loss-in-39-years/article69904116.ece> ; <https://www.thehindu.com/sci-tech/energy-and-environment/global-warming-pollution-are-stripping-vibrant->

colours-from-nature/article70614888.ece ; <https://indianexpress.com/article/explained/explained-climate/lakshadweep-coral-reefs-undergoing-severe-bleaching-9311488/>

Subject:) Environment

Topic:) Pollution

Subtopic:)

Q.78)

Ans) b

Exp) Option b is the correct answer.

Bioluminescence is the ability of living organisms to produce and emit light through a biochemical reaction. It is most common in marine ecosystems, though some terrestrial organisms also display it. It is important to clearly distinguish between 2 concepts that are bioluminescence **and biofluorescence**. **Bioluminescence is when light is produced chemically within the organism** and biofluorescence is when light is absorbed and re-emitted at a different wavelength by the organism.

Option I is correct. Marine dinoflagellates are one of the most common natural sources of bioluminescence in coastal waters. When waves disturb the water surface, these microscopic plankton activate a chemical reaction involving luciferin and luciferase, producing a characteristic **blue glow**, often referred to as a “blue tide.” This light production is defensive in nature, believed to startle predators or attract secondary predators to attack their attackers.

Option II is correct. Many jellyfish species possess bioluminescent capabilities. In several species, light emission results from specialised photoproteins or luciferin-luciferase reactions. The light may serve multiple ecological purposes such as predator deterrence, prey attraction and intraspecies communication. Bioluminescence is especially common in **deep-sea jellyfish**, where sunlight does not penetrate.

Option III is correct. Fireflies are classic terrestrial examples of bioluminescence. They produce light in abdominal light organs using luciferin and luciferase. Unlike marine organisms where light may be defensive, in fireflies the primary function is **sexual signalling and mate recognition**. The light produced is “cold light”, meaning almost no heat is generated which is a highly efficient biochemical process.

Option IV is incorrect. Termites do not possess any known biochemical mechanism for internal light production. Although some insects such as fireflies and glowworms are bioluminescent, termites are not.

Option V is incorrect. Certain **parrots** exhibit **biofluorescence**, not bioluminescence.

Biofluorescence occurs when an organism absorbs high-energy light (such as ultraviolet light) and re-emits it at a longer wavelength. This process requires an external light source. It does not involve luciferin-luciferase chemistry and does not produce light independently. Therefore, parrots do not generate light chemically and are not bioluminescent.

Source: <https://timesofindia.indiatimes.com/etimes/trending/10-glow-in-the-dark-insects-and-how-they-use-bioluminescence/articleshow/124156662.cms> ; <https://timesofindia.indiatimes.com/etimes/trending/glow-in-the-dark-beings-birds-of-paradise-emit-light-says-study/articleshow/123649131.cms> ; <https://www.thehindu.com/life-and-style/travel/kayaking-on-kochis-bioluminescent-backwaters/article69403315.ece> ; <https://indianexpress.com/article/upsc-current-affairs/upsc-essentials/knowledge-nugget-bioluminescence-blue-tide-upsc-filobolus-manipularis-9632097/> ; <https://indianexpress.com/article/cities/mumbai/juhu-beach-waves-blue-night-viral-sight-mumbai-science-10523827/>

Subject:) Environment

Topic:) Biodiversity

Subtopic:)

Q.79)

Ans) b

Exp) Option b is the correct answer.

Biodiversity conservation strategies are broadly classified into **in-situ** and **ex-situ** methods.

- **In-situ conservation** involves protecting species in their natural habitats.

- **Ex-situ conservation** involves conserving species outside their natural habitats under controlled or artificial conditions.

Option I is correct. **Botanical Gardens** are an **ex-situ** method. They conserve plant species outside their natural ecosystems. Rare and endangered plants are cultivated, preserved, and sometimes reintroduced into the wild. They also play an important role in research, education, and genetic conservation.

Option II is incorrect. **Biosphere Reserves** are an **in-situ** conservation method. They protect entire ecosystems within their natural settings. They include core, buffer, and transition zones to conserve biodiversity while allowing sustainable human activity.

Option III is correct. **Seed Banks** are **ex-situ** conservation method. They store seeds under controlled temperature and humidity conditions for long-term preservation of plant genetic material. They act as genetic insurance against extinction due to habitat loss, climate change, or disasters.

Source: <https://egyankosh.ac.in/bitstream/123456789/91214/1/Unit-11.pdf> - UNIT 11 - In-Situ and Ex-Situ Conservation - Ex-situ conservation - Page 234

Subject:) Environment

Topic:) Biodiversity

Subtopic:)

Q.80)

Ans) c

Exp) Option c is the correct answer.

The Climate and Clean Air Coalition (CCAC) has intensified global efforts to reduce **short-lived climate pollutants (SLCPs)**, often referred to as **super pollutants**, which have a disproportionately high warming impact in the near term. **At COP30 of UNFCCC**, it launched a targeted implementation mechanism to help developing countries embed coordinated policy and technical interventions to curb such emissions.

Statement I is correct. At COP30, the CCAC launched the **Super Pollutant Country Action Accelerator**, a three-year programme aimed at assisting ODA-eligible countries (Official Development Assistance of OECD eligible countries). The initiative provides:

- Embedded government coordination teams (4–7 staff members for three years)
- 4–5 tailored technical assistance projects per country
- Financial support of approximately USD 4–5 million per country

The programme seeks to reduce emissions across multiple super pollutant sectors and aims to scale up to 30 countries by 2030.

Statement II is incorrect. The Accelerator does **not** primarily focus on carbon dioxide (CO₂), nitrogen dioxide (NO₂), or sulphur hexafluoride (SF₆). Instead, it targets **short-lived climate pollutants** such as:

- **Methane (CH₄)**
- **Black carbon**
- **Hydrofluorocarbons (HFCs)**
- **Tropospheric ozone precursors**

These pollutants have high global warming potential in the short term and also significantly impact air quality, health, and agriculture. CO₂ reduction is handled under broader climate mitigation frameworks, while the CCAC specifically focuses on super pollutants with rapid climate and health benefits.

Source: <https://www.ccacoalition.org/content/launch-ccac-super-pollutant-country-action-accelerator> ; <https://www.ccacoalition.org/sites/default/files/resources/files/Supercharging%20Action%20on%20Super%20Pollutants.pdf>

Subject:) Environment

Topic:) Pollution

Subtopic:)

Q.81)

Ans) a

Exp) Option a is the correct answer.

India's Total Fertility Rate (TFR) has recently fallen to 1.9, below the replacement level of 2.1. In demographic analysis, the concept of **replacement level fertility** is crucial because it determines whether a population will grow, stabilize, or decline in the long run, assuming no migration.

Replacement level fertility refers to the **average number of children a woman must have to replace herself and her partner**, ensuring that one generation exactly replaces the previous one.

In most countries, this value is approximately 2.1, not exactly 2.0. The additional 0.1 accounts for **child mortality and the fact that not all children survive to reproductive age**. When TFR falls below this level, the population eventually begins to age and may decline, unless offset by migration.

Knowledge Base:

Crude Birth Rate (CBR) - the number of children born per 1,000 people in the population in a year.

Source: <https://www.thehindu.com/sci-tech/health/indias-infant-mortality-rate-touches-low-of-25-down-from-40-in-2013/article70011215.ece> ; <https://www.thehindu.com/news/national/indias-population-to-stabilise-by-2080-due-to-dip-in-fertility-rate-iasp/article70341029.ece>

Subject:) Economy

Topic:) Social Development

Subtopic:)

Q.82)

Ans) d

Exp) Option d is the correct answer.

In any modern economy, businesses require **capital** to expand operations, invest in new projects, repay debt or innovate. Broadly, capital can be raised through **internal accruals, borrowing (debt), or issuing ownership shares (equity)**. When companies raise funds by issuing equity shares to the public, they access the **capital market**. The capital market in India is regulated by the **Securities and Exchange Board of India (SEBI)**, which ensures transparency, investor protection, and fair practices.

Statement I is correct. An IPO is the **first public issue of equity shares by a private company**. Through this process, the company mobilises equity capital from the public and becomes listed on a recognised stock exchange. Post-listing, it must comply with continuous disclosure and corporate governance norms.

Statement II is correct. Grey Market Premium (GMP) indicates the **unofficial premium at which IPO shares trade before formal listing**. Importantly, the grey market operates **outside the regulatory framework** of SEBI and stock exchanges. Transactions are informal and carry legal risk, as there is no official oversight or investor protection mechanism.

Statement III is correct. The **Draft Red Herring Prospectus (DRHP)** is a preliminary disclosure document filed by a company with **SEBI** before launching an IPO. It provides detailed information on the company's **business model, financial statements, management profile, risk factors and pending litigations**. However, it does not disclose the final issue price of the IPO, since price discovery happens later.

Source: <https://investor.sebi.gov.in/pdf/reference-material/ppt/PPT-3%20How%20to%20invest%20in%20Initial%20Public%20Offer%20Updated%2030%20Sep%202022.pdf> ;
<https://www.nism.ac.in/understanding-drhp-rhp-and-prospectus/> ;
https://www.sebi.gov.in/sebi_data/faqfiles/may-2025/1747290561386.pdf ; <https://www.business-standard.com/about/what-is-gmp>

Subject:) Economy

Topic:) Capital Market

Subtopic:)

Q.83)

Ans) c

Exp) Option c is the correct answer.

The **Indian financial system** is broadly divided into the money market and the capital market based on the maturity period of financial instruments.

- The **money market deals with short-term instruments**, generally having a maturity of up to one year, and plays a crucial role in meeting the short-term liquidity requirements of governments, banks, financial institutions, and corporations. In India, the money market is regulated primarily by the RBI.
- Typical **money market instruments** include **Treasury Bills, Commercial Papers, Certificates of Deposit, Cash Management Bills, Collateralized Borrowing and Lending Obligations, repos**, and other short-duration liquidity instruments.

Option I is incorrect: State Development Loans (SDLs)

State Development Loans are **dated securities issued by State Governments** to finance their fiscal deficit. They generally have **medium- to long-term maturity** (generally 10 years) and are auctioned by the RBI.

They are part of the capital market, not the money market.

Option II is correct: Treasury Bills (T-bills)

These are **short-term debt instruments** issued by the Government of India in **three tenors: 91 days, 182 days, and 364 days**.

They are instruments issued at a discount to the face value and **form an integral part of the money market**.

Option III is correct: Tri-party Repo

Tri-party repo means a repo contract where a third entity (apart from the borrower and lender), called a Tri-Party Agent, acts as an intermediary between the two parties to the repo to facilitate services like collateral selection, payment and settlement, custody and management during the life of the transaction. Currently, the borrowing/lending under tri-party repo is facilitated for tenors up to next 7 days and for 3 months end date. A member can request for making available a particular tenor up to a period of 365 days. **Thus, as the maturity is less than 1 year, they are money market instruments.**

Earlier, Collateralised Borrowing and Lending Obligations (CBLO) was a short-term money market instrument operated by CCIL (now replaced by the Tri-party Repo (TREPS) mechanism).

Option IV is correct: Commercial Papers (CPs)

Commercial Papers are **unsecured short-term promissory notes issued by companies** to meet working capital needs. Their **maturity ranges from 7 days to one year**.

They are **money market instruments**.

Option V is correct: Cash Management Bills (CMBs)

CMBs are **short-term instruments issued by the Government of India** to meet temporary cash flow mismatches. Their **maturity is less than 91 days**.

Hence, **they are money market instruments**.

Option VI is correct: Repurchase Agreements (Repos)

A Repo is a **short-term borrowing mechanism** where securities are sold with an agreement to repurchase them later. They are key instruments used by the RBI for liquidity management in the money market. Repo transactions are **typically overnight or short-term**.

Repo is a core money market instrument.

Knowledge Base: **Capital Market**

The **capital market** is a segment of the financial system that deals with **medium- and long-term funds**, generally having a **maturity of more than one year**. Unlike the money market (which meets short-term liquidity needs), the capital market supports **long-term capital formation** in the economy.

Major Components of Capital Market:

1. Primary Market: Where new securities are issued for the first time (e.g., IPOs).

2. Secondary Market: Where existing securities are traded among investors.

Examples of Capital Market Instruments:

- Equity Shares (Stocks), Preference Shares, Debentures, Corporate Bonds, Government Securities (G-Secs) with maturity above one year and State Development Loans (SDLs)

Regulatory Authority:

The capital market in India is regulated by the **Securities and Exchange Board of India (SEBI)**.

Source:) <https://www.rbi.org.in/commonman/English/Scripts/FAQs.aspx?Id=711#30>

<https://www.ccilindia.com/triparty-repo> (Q.14)

Subject:) Economy

Topic:) Capital Market

Subtopic:)

Q.84)

Ans) b

Exp) Option b is the correct answer.

The **Reserve Bank of India** manages liquidity in the banking system and in the economy as a whole through multiple monetary policy instruments. These include **Open Market Operations (OMOs)**, **Cash Reserve Ratio (CRR)** adjustments, and **Repo/Reverse Repo operations under the Liquidity Adjustment Facility (LAF)**.

Open Market Operations (OMOs) specifically involve the outright purchase or sale of **Government Securities (G-Secs)** in the open market to manage durable liquidity. Not every liquidity adjustment tool used by RBI qualifies as an OMO.

Option I is correct. When RBI **sells Government Securities** in the open market, it absorbs liquidity from the banking system/economy. This is a classic example of an **Open Market Operation**, aimed at reducing excess money supply on a durable basis.

Option II is incorrect. Reduction in the **Cash Reserve Ratio (CRR)** for commercial banks by RBI changes the proportion of deposits banks must keep with RBI. While this affects liquidity, it is a **reserve requirement tool**, not an OMO. It does not involve buying or selling securities in the market.

Option III is correct. When RBI **purchases Government Securities** from the secondary market, it injects liquidity into the system. This is an outright transaction in G-Secs and therefore qualifies as an **Open Market Operation**.

Option IV is incorrect. The **Repo Rate** is the rate at which RBI lends short-term funds to banks against securities. It is a **policy rate tool**, not a market purchase/sale of securities. Hence, it is **not an OMO**.

Source: <https://www.rbi.org.in/commonman/english/scripts/FAQs.aspx?Id=711#5> ;

<https://egyankosh.ac.in/bitstream/123456789/37982/1/Unit-14.pdf> - UNIT 14 - Credit and Monetary Policy In India - RBI's Intervention In The Money Market: Refinance, Marginal Standing Facility And Liquidity Adjustment Facility – Page 21

Subject:) Economy

Topic:) Monetary Policy

Subtopic:)

Q.85)

Ans) b

Exp) Option b is the correct answer.

Inflation refers to a **persistent increase in the general price level** of goods and services in an economy. It is important to understand that inflation is not about the rise in the price of a single commodity. It concerns a **sustained rise in average prices across the economy**. Inflation may arise because **aggregate demand becomes excessive** or because **costs of production increase**.

Pair I is correct. When **government expenditure rises sharply** and the economy is already close to full employment, total demand in the economy increases beyond its productive capacity. **Producers cannot increase output** further because

most resources are already fully utilised. As a result, prices start rising. This situation is known as **demand pull inflation** because the excess demand pulls prices upward.

Pair II is correct. A sudden rise in **crude oil prices**, wages, or raw material costs increases the **cost of production**. Producers often pass on this increased cost to consumers by raising prices. This type of inflation originates from the **supply side of the economy** and is known as **cost push inflation**. The essential concept here is **rising production costs leading to higher prices**.

Pair III is incorrect. When there is a **continuous decline in the general price level**, accompanied by **falling output and rising unemployment**, the economy is experiencing **deflation**. Deflation usually results from **weak aggregate demand**. Businesses reduce production because goods remain unsold, which further reduces income and employment. The defining feature of deflation is a **sustained fall in prices along with economic slowdown**.

Pair IV is incorrect. **Disinflation (not hyper-inflation)** refers to a **gradual reduction in the rate of inflation**. Prices continue to rise, but at a **slower rate than before**. For example, if inflation falls from six percent to four percent, it is disinflation. It is not deflation because **prices are still increasing but at a slower rate than earlier**. Policymakers often aim for gradual disinflation to avoid a sudden spike in unemployment. The key phrase to remember is **decline in inflation rate, not decline in price level**.

Source: <https://egyankosh.ac.in/bitstream/123456789/73889/1/Block-3.pdf> - UNIT 7 - Inflation: Concept, Types and Measurement - Types of Inflation - Page 77

Subject:) Economy

Topic:) Inflation

Subtopic:)

Q.86)

Ans) a

Exp) Option a is the correct answer.

The **National Air Quality Monitoring Programme (NAMP)** is **implemented by the Central Pollution Control Board (CPCB)** to systematically monitor ambient air quality at different locations across India.

Under NAMP, **four air pollutants** have been identified for regular monitoring at all locations:

- **Sulphur Dioxide (SO₂) (Option I is correct)**
- **Oxides of Nitrogen (as NO₂) (Option V is correct)**
- Respirable Suspended Particulate Matter (RSPM / PM₁₀)
- Fine Particulate Matter (PM_{2.5})

Other pollutants in the given list, such as **Carbon Monoxide (CO), Ammonia (NH₃), and Lead (Pb)**, are **not part of the core pollutants monitored under NAMP at all locations**, but they are included among the 12 pollutants listed under the **National Ambient Air Quality Standards (NAAQS)**.

Hence, **only SO₂ and NO₂** from the given list are **identified for regular monitoring under NAMP**.

Source:) <https://cpcb.nic.in/about-namp/>

https://cpcb.nic.in/upload/NAAQS_2019.pdf

Subject:) Environment

Topic:) Pollution

Subtopic:)

Q.87)

Ans) b

Exp) Option b is the correct answer.

The **Balance of Payments (BoP)** is a systematic **record of all economic transactions between the residents of a country and the rest of the world during a given period** (usually a year).

It is broadly **divided into two main accounts**:

- **Current Account** – Records transactions relating to:
 - **Trade in goods** (exports and imports)
 - **Trade in services** (IT, tourism, banking, etc.)
 - **Income** (interest, dividends)
 - **Current transfers** (remittances, gifts, grants)
- **Capital Account** – Records transactions that:
 - Create or reduce financial assets and liabilities
 - Include **foreign investments, external borrowings, loan repayments, and banking capital flows.**

Thus, while the Current Account reflects income and expenditure flows, the Capital Account reflects changes in ownership of assets and liabilities between residents and non-residents.

Pair I is incorrect: When a foreign tourist purchases Indian handicrafts, it is treated as **export of goods (merchandise trade)**. Exports and imports of goods are recorded under the **Current Account of the Balance of Payments**. This is a **"visible" trade item** and is recorded in the Current Account.

Pair II is correct: Provision of IT services by an Indian company to a German firm constitutes **export of services**. The export of software services is a classic example of **"Invisibles."** Trade in services forms part of the **Current Account**.

Pair III is correct: Issuing sovereign bonds to foreign investors results in **capital inflow in the form of external borrowing**, creating a liability for India. Such borrowings are recorded under the **Capital Account**.

Pair IV is incorrect: Repayment of the principal amount of an external commercial borrowing reduces India's external liability. This is a **capital outflow** and is **recorded under the Capital Account, not the Current Account**.

(Only interest payments would be recorded under the Current Account.)

Source:) <https://egyankosh.ac.in/bitstream/123456789/23157/1/Unit-7.pdf> Page: 3.

https://mospi.gov.in/sites/default/files/Statistical_year_book_india_chapters/Chapter%20No.4.pdf

Subject:) Economy

Topic:) External Sector

Subtopic:)

Q.88)

Ans) d

Exp) Option d is the correct answer.

Option d is correct: **Agentic AI** refers to a **class of advanced Artificial Intelligence systems** that **function as autonomous "agents."**

- Such systems are capable of **perceiving their environment, setting goals, making decisions, and executing a sequence of actions with minimal human intervention.**
- Such systems can plan and carry out **multi-step tasks**, interact with digital tools or software environments, monitor outcomes, and adjust their actions to achieve a specified objective.
- **Unlike conventional AI systems** that merely **respond to prompts** or perform narrowly defined tasks, **Agentic AI demonstrates goal-directed behaviour, adaptive reasoning, and iterative problem-solving, with minimal continuous human supervision.**

Option a is incorrect: **Explainable AI (XAI):**

This refers to **AI systems whose decision-making processes are transparent, interpretable, understandable and reliable to humans.** The focus is on understanding how an AI arrives at a result, not on autonomous goal-driven action.

Option b is incorrect: **Generative AI:**

This category of AI focuses on **creating new content** such as text, images, audio, or code based on patterns learned from training data (e.g., large language models or image generators).

It primarily **generates outputs in response to prompts rather than autonomously planning and executing complex tasks.**

Option c is incorrect: **Edge AI:**

This refers to the **deployment of AI models directly on edge devices** such as smartphones, IoT sensors, or embedded systems, enabling local data processing with reduced reliance on cloud computing. It concerns where AI runs, not autonomous decision-making capability.

Source:) <https://www.ibm.com/think/topics/ai-agents>

<https://www.thehindu.com/business/steering-intelligence-building-governance-foundations-for-the-agentic-ai/article70427350.ece>

<https://indianexpress.com/article/technology/artificial-intelligence/ai-agents-who-is-building-what-openai-google-krutrim-10070144/>

Subject:) Science and Technology

Topic:) Technology

Subtopic:)

Q.89)

Ans) b

Exp) Option b is the correct answer.

Statement I is incorrect: A Central Public Sector Enterprise (CPSE) that has **made profits continuously for the last three years** and has a **positive net worth** becomes eligible for Miniratna classification (**profit in three years out of last five years is not a criteria either for Miniratna or Navratna company**).

For Navratna status, the requirements are more stringent. The eligibility criteria for a CPSE to attain a Navratna status is as follows:

- **Navratna Eligibility:** To be considered for Navratna status, a CPSE must already be a **Miniratna Category-I** and a **Schedule 'A'** company.
- **Performance Score:** It must have obtained an **'Excellent'** or **'Very Good'** rating in its Memorandum of Understanding (MoU) for **three of the last five years**.
- **Composite Score:** It must also achieve a composite score of **60 or above** out of 100 based on six performance indicators:
 - Net Profit to Net Worth
 - Manpower Cost to Total Cost of Production/Services
 - Profit before depreciation, interest and taxes to capital employed,
 - Profit before interest and taxes to turnover,
 - Earnings Per Share (EPS)
 - Inter-Sectoral Performance

Statement II is correct: The Government of India has recently granted Navratna status to:

- **Indian Railway Catering and Tourism Corporation (IRCTC)**
- **Indian Railway Finance Corporation (IRFC)**
- **Numaligarh Refinery Limited (NRL)**

With the upgradation of IRCTC and IRFC, an important development in the railway sector is that:

Out of **12 Central Public Sector Enterprises (CPSEs) in the railway sector**,

- 7 are listed companies, and
- **All 7 listed railway CPSEs now have Navratna status.**

This marks a significant strengthening of financial and operational autonomy within the railway PSU ecosystem.

Source:) <https://www.thehindu.com/business/Industry/numaligarh-refinery-accorded-navaratna-status-informs-petroleum-minister-hardeep-puri/article70349886.ece>

<https://www.thehindu.com/news/national/irctc-irfc-upgraded-to-navratna-status-railway-minister-compliments/article69285436.ece>

<https://www.pib.gov.in/newsite/PrintRelease.aspx?relid=107091®=3&lang=2>

<https://indianexpress.com/article/explained/explained-economics/navratna-status-irctc-irfc-9868272/>

Subject:) Economy

Topic:) Sectors of Economy

Subtopic:)

Q.90)

Ans) c

Exp) Option c is the correct answer.

The **Payments Regulatory Board (PRB)** is the **apex regulatory and statutory body** responsible for the **regulation and supervision of payment and settlement systems in India**. It functions within the institutional framework of the Reserve Bank of India (RBI) and derives its authority from the **Payment and Settlement Systems Act, 2007**.

Statement I is correct: The **PRB is headed by the Governor of the Reserve Bank of India (RBI)**, who serves as the **ex-officio Chairperson**. This ensures that the regulation of payments remains closely integrated with the broader monetary and banking supervision handled by the RBI.

The **composition of the PRB includes members from both the RBI and nominees of the Central Government**. Typically, the Board consists of:

- The **RBI Governor (Chairperson)**
- A **Deputy Governor of RBI**
- One additional RBI official nominated by the RBI
- **Three nominees of the Central Government**

This composition aims to ensure coordination between the central bank and the government in regulating the rapidly expanding digital payment ecosystem.

Statement II is correct: Under the **Payment and Settlement Systems Act, 2007**, the **PRB is entrusted, on behalf of the Reserve Bank of India, with the regulation and supervision of payment and settlement systems in India**, including authorisation of payment system operators and oversight of system safety and efficiency.

Source:) <https://www.thehindu.com/business/rbi-constitutes-payments-regulatory-board-headed-by-governor/article70113688.ece>

<https://www.thehindubusinessline.com/money-and-banking/rbi-governor-to-chair-new-payments-regulatory-board/article70113704.ece> "

<https://www.rbi.org.in/commonman/Upload/English/FAQs/PDFs/FAQ20102025FUL.pdf>

Subject:) Economy

Topic:) Monetary Policy

Subtopic:)

Q.91)

Ans) b

Exp) Option b is the correct answer.

Waste-to-Energy (WtE) technologies consist of any waste treatment process that creates energy in the form of electricity or heat from several types of waste: from the semi-solid (e.g. thickened sludge from effluent treatment plants) to liquid (e.g. domestic sewage) waste. For example, Waste-to-Energy (WtE) technologies may refer to a group of processes that **convert municipal solid waste (MSW) into usable forms of energy**, such as electricity, heat, biogas, or synthetic fuels.

WtE technologies broadly fall into **two categories**:

- **Thermal Technologies** – Incineration, Gasification, Pyrolysis
- **Biological Technologies** – Anaerobic Digestion (Biomethanation)

Option I is correct: Incineration

Incineration involves the **controlled combustion of municipal solid waste at high temperatures (750–1100°C) in the presence of oxygen**.

The heat generated is used to produce steam, which drives turbines for electricity generation.

It significantly reduces waste volume (up to 80–90%) and is **one of the most widely adopted Waste-to-Energy technologies globally**.

Option II is correct: Gasification

Gasification is a thermo-chemical process in which carbon-based **waste is heated at high temperatures with limited oxygen**.

Instead of complete combustion, it **produces synthetic gas (syngas)** consisting mainly of carbon monoxide and hydrogen. **Syngas can be used for electricity generation** or converted into fuels and chemicals.

Hence, gasification is also an **example of Waste-to-Energy technology**.

Option III is incorrect: Vermicomposting

Vermicomposting is a biological process where **earthworms decompose organic waste into nutrient-rich compost (vermicast)**.

While it is excellent for soil health and recycling nutrients, it does not generate electricity, heat, or fuel.

Therefore, **it is a waste management technique but not a Waste-to-Energy technology**.

Option IV is correct: Pyrolysis

Pyrolysis is the **thermal decomposition of waste in the absence of oxygen**.

It produces pyrolysis oil, combustible gases, and char, all of which can be used as fuels or energy sources.

Hence, **pyrolysis is classified as a WtE technology**.

Option V is incorrect: Phytoremediation

Phytoremediation **uses plants to remove, stabilize, or detoxify pollutants** from soil, water, or waste sites.

It is an **environmental remediation technique** aimed at pollution control, not at energy recovery.

No energy is produced in this process. Hence, **it is not a WtE technology**.

Source:) <https://unfccc.int/technology/waste-to-energy-technologies>

<https://www.thehindu.com/sci-tech/energy-and-environment/what-is-waste-to-energy-technology/article70577241.ece>

<https://mnre.gov.in/en/waste-to-energy/>

<https://www.thehindu.com/sci-tech/health/taking-health-into-consideration-is-garbage-incineration-the-way-forward-to-tackle-waste/article68908365.ece>

<https://eerem.delhi.gov.in/eerem/waste-energy>

Subject:) Science and Technology

Topic:) Energy

Subtopic:)

Q.92)

Ans) a

Exp) Option a is the correct answer.

Statement I is correct: Hypersonic Glide Vehicles (HGVs) are indeed **much harder to intercept than conventional ballistic missiles**.

- **Traditional missile defence systems** are designed to **track ballistic missiles that follow predictable parabolic paths**.
- HGVs, however, complicate interception because of their **high speed** (above Mach 5), **low-altitude flight**, and **unpredictable maneuvers**. This makes detection, tracking, and interception extremely challenging.

Statement II is correct: Ballistic missiles generally **follow a fixed and predictable parabolic trajectory** after launch. This allows defence systems to calculate their path and determine likely interception points.

In contrast, **HGVs do not follow a parabolic trajectory**. They **re-enter the atmosphere** earlier and at much **lower altitudes**. They can subsequently **glide to their target** while executing advanced evasive **manoeuvres under guided flight**.

Link Between Statement II and Statement I:

Statement II directly explains why Statement I is true. The maneuverability and lower-altitude flight path of HGVs (Statement II) are the reasons they are significantly harder to intercept compared to ballistic missiles (Statement I).

Hence, **Both Statement I and Statement II are correct and Statement II explains Statement I**.

Source:) <https://www.orfonline.org/expert-speak/how-hypersonic-weapons-are-redefining-warfare>
<https://www.thehindu.com/sci-tech/technology/explained-the-arms-race-towards-hypersonic-weapons/article37874968.ece>
<https://indianexpress.com/article/explained/drdo-hypersonic-glide-missile-republic-day-parade-10495213/>
<https://www.thehindu.com/news/national/understanding-hypersonic-weapons/article65249442.ece>
<https://swarajyamag.com/defence/explained-what-are-hypersonic-missiles-and-why-countries-with-nukes-need-them>
 Subject:) Science and Technology
 Topic:) Defence
 Subtopic:)

Q.93)

Ans) d

Exp) Option d is the correct answer.

Option d is correct: A **Scramjet** (Supersonic Combustion Ramjet) is a type of **air-breathing propulsion system designed to operate at hypersonic speeds** (generally above Mach 5). Unlike conventional rocket engines, a **Scramjet does not carry an onboard oxidizer**; instead, **it intakes oxygen from the atmospheric air during flight**.

- The working principle is based on **ram compression**: as the vehicle moves at very high speeds, incoming air is compressed naturally without the need for a mechanical compressor.
- Fuel (usually hydrogen or hydrocarbon) is injected into this high-speed airflow, and combustion occurs supersonically.
- Its unique design makes it ideal for **hypersonic flight** and potential space-access applications.

Option a is incorrect: Solid Propellant Rocket Engine

It **carries both fuel and oxidizer mixed together in solid form** within the combustion chamber.

It does not depend on atmospheric oxygen and hence is **not air-breathing**.

Option b is incorrect: Liquid Propellant Rocket Engine

It **stores liquid fuel and liquid oxidizer separately** in tanks and pumps them into the combustion chamber.

Since it carries its own oxidizer, **it is not an air-breathing system**.

Option c is incorrect: Cryogenic Engine

It is a type of liquid rocket engine that uses **cryogenic fuels** (e.g., liquid hydrogen) and a **cryogenic oxidizer** (e.g., liquid oxygen).

As it carries onboard oxidizer, **it is also not air-breathing**.

Source:) <https://www.thehindu.com/sci-tech/science/isro-conducts-experimental-flight-of-air-breathing-propulsion-system/article68432854.ece>

<https://www.thehindu.com/sci-tech/science/drdo-achieves-major-breakthrough-in-hypersonic-weapon-technology/article69492050.ece>

Subject:) Science and Technology

Topic:) Technology

Subtopic:)

Q.94)

Ans) b

Exp) Option b is the correct answer.

Radio Frequency Identification (RFID) is a **wireless communication technology** that **uses radio waves** to identify, track, and exchange data with tagged objects automatically.

A **key advantage of RFID over barcodes** is that it **does not require direct line-of-sight** and can **read multiple tags simultaneously**.

RFID technology is widely used in toll collection, supply chain management, access control systems, smart transit cards, livestock tracking, and asset management.

Option I is correct: Electronic toll collection systems such as FASTag

FASTag operates on passive RFID (Radio Frequency Identification) technology. A tag affixed on the vehicle's windshield communicates with RFID readers installed at toll plazas through radio waves, enabling automatic vehicle identification and electronic toll deduction without physical contact.

Option II is incorrect: Biometric authentication in Aadhaar using fingerprint scanning

Aadhaar authentication through fingerprint scanning relies on biometric sensors that capture fingerprint data and match it with stored biometric records in a central database.

This process does not involve radio frequency tags or wireless identification technology; hence **RFID is not used**.

Option III is correct: Inventory tracking and supply chain management in warehouses

RFID tags attached to goods or pallets store electronically coded information. RFID readers can scan multiple items simultaneously without direct line-of-sight, enabling real-time tracking, automated stock management, reduced pilferage, and improved logistics efficiency.

Option IV is correct: Contactless metro and smart transit cards

Contactless smart cards used in metro systems function on RFID/NFC (Near Field Communication) technology. When the card is brought near a reader, it communicates wirelessly via radio waves to authenticate and deduct fare, eliminating the need for swiping or physical insertion.

Option V is incorrect: Magnetic stripe-based debit and credit cards

Magnetic stripe cards store data in magnetized stripes and require physical swiping through a reader. **They do not use radio frequency signals for communication.** (Only contactless EMV cards use RFID/NFC technology, not magnetic stripe cards.)

Source:) <https://indianexpress.com/article/explained/everyday-explainers/explained-what-are-rfid-tags-that-will-be-used-to-track-amarnath-yatra-pilgrims-jammu-kashmir-7923842/>

<https://indianexpress.com/article/upsc-current-affairs/upsc-essentials/knowledge-nugget-rfid-technology-fastag-key-concept-upsc-exam-10076365/>

<https://m.economictimes.com/small-biz/sme-sector/rfid-technology-can-help-in-efficient-supply-chain-management-bar-code-india/articleshow/105084684.cms>

Subject:) Science and Technology

Topic:) Technology

Subtopic:)

Q.95)

Ans) b

Exp) Option b is the correct answer.

Anti-Submarine Warfare – Shallow Water Craft (ASW-SWC) are specialized **naval vessels designed to detect, track, and neutralize enemy submarines in coastal and shallow waters.** These ships are smaller, highly maneuverable, and equipped with **sonar, torpedoes, depth charges, and anti-submarine rockets,** making them **effective in shallow waters** where larger warships or submarines cannot operate efficiently.

Option I is correct: INS Mahe

INS Mahe is an Anti-Submarine Warfare–Shallow Water Craft (ASW-SWC) of the Indian Navy. It belongs to the **Mahe class,** which comprises indigenously designed and built shallow-water corvettes. It is **first of the indigenously designed and built Mahe-class ASW-SWC.**

- **Designed and constructed by Cochin Shipyard Limited,** Kochi, INS Mahe is the lead ship of eight vessels in her class.
- It is specially designed to undertake anti-submarine operations in coastal and shallow waters.

Option II is correct: INS Anjadip

INS Anjadip is another ASW-SWC of the Indian Navy. It is **part of the Arnala-class** and has been built by **Garden Reach Shipbuilders & Engineers (GRSE), Kolkata.**

- It is specifically designed to address the challenges of the littoral combat environment—the **coastal and shallow waters** vital for the nation's security.
- The vessel is engineered to act as a '**Dolphin Hunter**', focused on the detection, tracking, and neutralisation of enemy submarines in coastal areas.

Option III is incorrect: INS Vikramaditya

INS Vikramaditya is **India's flagship aircraft carrier, not an ASW vessel.** It serves as a '**floating sovereign airfield**', providing a launch platform for a wide array of aircraft, including MiG-29K fighter jets, MH60R, Kamov, Sea King, Chetak and ALH helicopters.

Option IV is incorrect: INS Arihant

INS Arihant is a nuclear-powered ballistic missile submarine (SSBN). It is **India's first indigenous** nuclear-powered submarine.

Commissioned in 2016, it is designed for strategic deterrence with nuclear ballistic missiles. It is part of India's nuclear triad.

Source:) <https://www.pib.gov.in/PressReleasePage.aspx?PRID=2190563®=3&lang=2>

<https://www.newsonair.gov.in/indian-navy-inducts-anjadip-fourth-anti-submarine-warfare-shallow-water-craft/>

<https://www.pib.gov.in/PressReleasePage.aspx?PRID=2231754®=3&lang=1> "

<https://timesofindia.indiatimes.com/defence/news/arihant-arighaat-aridhaman-inside-indias-growing-nuclear-submarine-triad-against-china-pakistan/articleshow/128710555.cms>

<https://indianexpress.com/article/india/twin-cbg-operation-ins-vikramaditya-vikrant-lead-navys-mega-ops-in-arabian-sea-8656455/>

Subject:) Science and Technology

Topic:) Defence

Subtopic:)

Q.96)

Ans) b

Exp) Option b is the correct answer.

Quick Response (QR) Codes are two-dimensional (2D) matrix barcodes that store information in a grid of black and white squares. **Developed in 1994 by the Japanese company Denso Wave**, they were initially designed for tracking automotive parts but are now widely used for digital payments, ticketing, authentication, inventory management, and information sharing.

Statement I is correct: QR Codes (Quick Response Codes) are **two-dimensional matrix codes** made of black and white square modules arranged in a grid. It is a two-dimensional machine-readable unique pattern that contains information on a specific task.

Unlike traditional barcodes that store information **only horizontally**, **QR codes encode data both horizontally and vertically**, which significantly **increases their data storage capacity**.

Statement II is correct: QR codes incorporate error-correction mechanisms (based on **Reed-Solomon error correction**). Because of this feature, **scanners can still retrieve the encoded information** even if a **portion of the code is damaged**, dirty, or partially obscured.

This makes QR codes more robust for real-world uses such as product tracking, digital payments, and ticketing.

Statement III is incorrect: A QR code does not require an internet connection to scan the code or decode the stored information. The **encoded data** (such as text or a URL) is stored directly within the QR pattern and **can be read offline** by a scanner or smartphone camera.

However, if the QR code contains a web link, then an internet connection will be required after scanning to access the webpage; not to decode the code itself.

Statement IV is correct: “Quishing” (QR phishing) is a form of cyber-fraud where attackers use malicious QR codes to redirect users to phishing websites or fraudulent payment portals.

This can happen when criminals:

- Replace legitimate QR codes with fake ones on posters, parking meters, or payment stands.
- Send QR codes via emails or messages that lead to credential-stealing websites.

Because QR codes are easy to generate and copy, users may unknowingly scan a fraudulent code.

Source:) <https://www.thehindu.com/sci-tech/technology/why-qr-codes-matter-and-how-to-use-them-effectively/article69368006.ece>

<https://www.thehindu.com/business/new-generation-barcodes-to-go-live-from-2027-gs1/article69594762.ece>

https://www.thehindu.com/sci-tech/technology/new-qr-code-to-have-colours-will-hold-more-data-says-japanese-inventor-hara/article66440329.ece#google_vignette

<https://www.thehindu.com/news/national/kerala/police-issue-warning-against-fake-qr-codes-in-public-places-in-kozhikode/article70066421.ece>

Subject:) Science and Technology

Topic:) Technology

Subtopic:)

Q.97)

Ans) a

Exp) Option a is the correct answer.

A **Zero-Knowledge Proof (ZKP)** is a **cryptographic protocol** in which **one party** (called the prover) **proves to another party** (called the verifier) that a **particular information/statement is true without revealing any information** other than the fact that the statement is true. In simple terms, it enables verification without disclosure.

For **example**, a person can prove that they are above 18 years of age without revealing their actual date of birth or identity details.

PUTTING THE 'PROOF' IN PRIVACY

- Posidex Technologies introduces zero knowledge proof blockchain technology in India
- ZKP technology is currently being developed by only a few countries like the US, Israel
- It allows processing of data at the host level without sharing or shifting the data itself
- They are cryptographic methods used to prove knowledge without revealing the data



- ZKP technology has been validated by a reputable global bank and tested successfully
- Current technologies limit cross-border data access, posing challenges for entities
- ZKP simplifies access to encrypted data while maintaining privacy and security
- Data residency legislation and regulations like GDPR restrict data sharing and migration
- Data protection laws in India further complicate data exchange

Knowledge Base:

Europe's Efforts to Protect Children Online

The **European Union** has been exploring the use of privacy-preserving technologies like Zero-Knowledge Proofs for **age-verification systems** on digital platforms. Under initiatives linked to the **Digital Services Act (DSA)**, the EU is working on systems that allow platforms to verify whether a user is **above a certain age (such as 18)** without requiring disclosure of personal information such as identity documents or exact birth date.

Source:) <https://www.thehindu.com/sci-tech/technology/what-is-the-european-commissions-age-verification-plan-to-protect-children-online-explained/article69826633.ece>

<https://www.thehindu.com/news/national/telangana/personal-data-can-be-shared-without-breaching-the-privacy-wall/article68344062.ece>

<https://timesofindia.indiatimes.com/city/hyderabad/hyderabad-based-firm-introduces-zero-knowledge-proof-blockchain-technology/articleshow/110346516.cms>

Subject:) Science and Technology

Topic:) Technology

Subtopic:)

Q.98)

Ans) a

Exp) Option a is the correct answer.

The **Willow quantum processor** has been **developed by Google**, specifically by its quantum computing division, **Google Quantum AI**.

It is a **superconducting qubit-based quantum chip** designed to perform highly complex computations that are practically infeasible for classical supercomputers. **Willow was reported to have achieved the first “verifiable quantum advantage”**, meaning:

- It **performed a specific computational task exponentially faster than classical systems**, and
- The **correctness of the quantum computation could be independently verified**.

This claim marks an important milestone in quantum computing research, distinguishing it from earlier demonstrations where speed advantage was shown but independent verification was more difficult.

Source:) <https://www.thehindu.com/sci-tech/science/what-is-googles-quantum-advantage-leap-explained/article70202625.ece>

<https://www.thehindu.com/sci-tech/science/google-claims-first-verifiable-quantum-advantage-for-willow-chip/article70191045.ece>

Subject:) Science and Technology

Topic:) Technology

Subtopic:)

Q.99)

Ans) c

Exp) Option c is the correct answer.

The **Wetland City Accreditation (WCA)** is an international **recognition under the Ramsar Convention on Wetlands**. It is awarded to cities that demonstrate effective conservation, sustainable management, and **wise use of urban and peri-urban wetlands**, while also promoting public awareness and community participation in wetland protection.

In 2025, Indore (Madhya Pradesh) and Udaipur (Rajasthan) became the first Indian cities to receive the Wetland City Accreditation.

- Indore** was recognised for conservation efforts around wetlands such as **Sirpur Lake**, which supports migratory birds and urban biodiversity.
- Udaipur**, often called the **City of Lakes**, was recognised for its conservation of interconnected urban lake systems (like **Pichola and Fateh Sagar**), sustainable tourism management, and long-term wetland protection initiatives.

Knowledge base:

Wetlands of Bhopal

Bhopal, often called the **“City of Lakes,”** has several important wetlands that play a crucial role in water supply, biodiversity conservation, and urban ecology.

- **Bhojtal (Upper Lake):** One of the oldest man-made lakes in India, built in the 11th century by Raja Bhoj. I

- **Lower Lake:** Located downstream of Bhojtal, it forms a connected lake system and supports urban ecological balance.
- **Bhoj Wetland:** The combined ecosystem of Upper and Lower Lakes is designated as a **Ramsar Site (Wetland of International Importance)** due to its ecological significance and role in supporting migratory birds.

Wetlands of Srinagar

Srinagar in Jammu & Kashmir has several natural wetlands associated with the **Kashmir Valley lake system**, which are vital for flood regulation, fisheries, and bird habitats.

- **Dal Lake:** A famous urban lake known for **floating gardens (Raad cultivation)** and houseboats; it also supports fisheries and tourism.
- **Hokersar Wetland:** A **Ramsar Site** and an important **bird sanctuary**, serving as a major wintering ground for migratory birds from Central Asia.
- **Anchar Lake:** Located near Srinagar and hydrologically linked with Dal Lake.

Source:) <https://www.ramsar.org/our-work/activities/wetland-city-accreditation>

<https://timesofindia.indiatimes.com/india/indore-and-udaipur-become-the-first-two-indian-cities-to-join-the-global-list-of-wetland-accredited-cities/articleshow/117558751.cms>

Subject:) Environment

Topic:) Wetland

Subtopic:)

Q.100)

Ans) d

Exp) Option d is the correct answer.

The **Economic Capital Framework (ECF)** refers to the **framework used by the Reserve Bank of India to determine the level of capital reserves (risk buffers) it should maintain** and the **amount of surplus that can be transferred to the Government of India**.

The RBI faces various financial risks such as currency fluctuations, interest-rate movements, and market volatility arising from its monetary operations and management of foreign exchange reserves. To safeguard against these risks, the RBI maintains **capital buffers** known as the **Contingency Risk Buffer (CRB)** and other provisions.

The Economic Capital Framework provides a systematic method to:

- **assess the minimum capital and risk buffers the RBI must maintain**, and
- **decide how much of its annual surplus (profit) can be transferred to the Government of India** after maintaining adequate reserves.

The present framework was formalized on the basis of recommendations of the **Bimal Jalan Committee (2019)**, which reviewed the RBI's economic capital requirements and suggested an appropriate range for the Contingency Risk Buffer.

Source:) <https://www.thehindu.com/business/Economy/finance-ministry-looking-into-how-to-enhance-dividends-from-rbi/article69582700.ece>

<https://www.thehindu.com/business/Economy/where-does-the-rbis-surplus-come-from-explained/article69615246.ece>

<https://www.thehindu.com/business/Economy/rbi-gives-269-lakh-crore-dividend-to-central-govt-for-fy25/article69610382.ece>

<https://economictimes.indiatimes.com/news/economy/finance/new-capital-framework-gives-rbi-more-room-to-handle-its-finances/articleshow/121473441.cms?from=mdr>

Subject:) Economy

Topic:) Monetary Policy

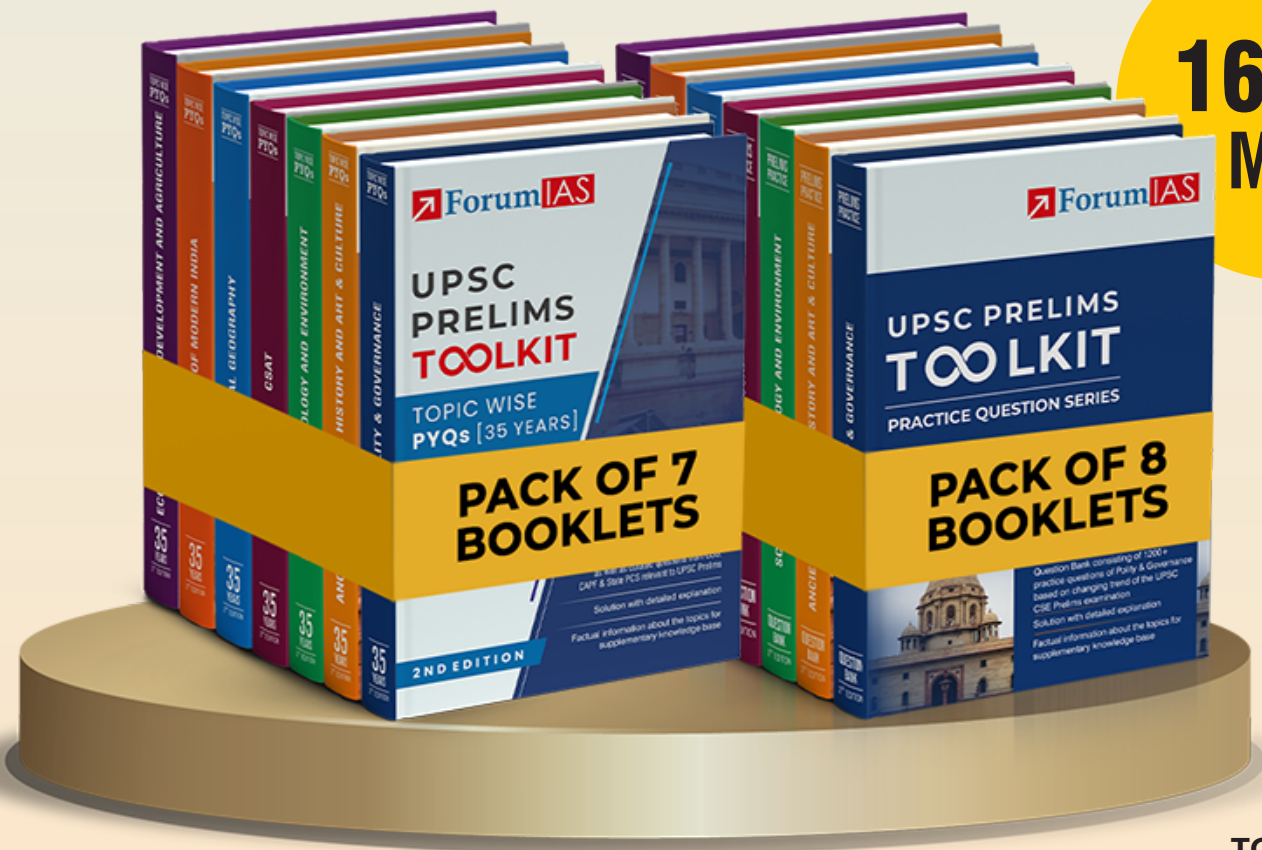
Subtopic:)

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